The Rise of H-2A: Causal Evidence on Guestworker Substitution for Settled Farm Labor *

Aakansha Jain

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Abstract

Where do farms turn when immigration enforcement thins their workforce? Non-citizen immigrant workers have long supplied most seasonal farm labor in the US. Two thirds of them lack work authorization, and immigration enforcement has contributed to a decline in these settled non-citizen workers since 2015. Meanwhile, the H-2A program, which lets farms legally hire seasonal foreign workers, has grown more than fourfold since 2008. This paper asks whether the decline in settled non-citizen farm workers contributed to the rise of the H-2A program, and how many H-2A workers farms hire for each worker lost. Using a panel of 740 commuting zones from 2008 to 2024, I instrument the contemporaneous settled non-citizen share of farm labor with its 2000 baseline share interacted with national enforcement shifts. I find that farms hire about one H-2A worker for each settled worker lost. Since H-2A contracts are shorter, this replaces 0.7 of the lost labor in full-time equivalents. The substitution accounts for 29 percent of aggregate H-2A growth since 2008. The effect appears only in crops, where the H-2A program is available, and not in year-round livestock. Under enforcement pressure, US farms are moving toward hiring through costlier legal channels with shorter contracts.
JEL codes: J43, J61, J68, K37, Q12.

Keywords: H-2A program; guestworkers; immigration enforcement; farm labor; shift-share instruments; substitution.

*Jain: Department of Economics, Iowa State University (email: aakansha@iastate.edu). I thank my advisors, John Winters and Peter Orazem, and my committee members, Rajesh Singh, Donghyuk Kim, and Biswa Das, for guidance and comments. All errors are my own.

1 Introduction

The U.S. hired farm workforce can be divided into three parts: domestic workers, settled non-citizen workers, and seasonal guestworkers hired through the H-2A visa program. Domestic workers comprise U.S.-born and naturalized citizens, and their numbers have remained largely stable, with some increase between 2014 and 2018. Settled non-citizen workers, who have long been the largest source of seasonal farm labor, are workers who have established residence in the United States rather than arriving for the season, and a large share of them lack work authorization. Their numbers have declined since 2015. Finally, H-2A guestworkers are similar to settled non-citizen labor, with the difference that they are hired legally and for a short period, averaging 5.7 months. H-2A certifications grew from about 87,000 positions in 2008 to 385,000 in 2024, more than fourfold. Figure 1 depicts these trends: H-2A rises and settled non-citizen labor falls while the total remains stable. The two series measure different things. H-2A positions are annual certifications, each covering under six months of work, while the ACS series count workers employed at the time of the survey. They are not comparable in levels, and the crossover should not be read as guestworkers outnumbering settled workers.

This paper asks whether the decline in settled farm workers contributed to the rise of the H-2A program, and the answer is not obvious. The two series can move this way for reasons unrelated to substitution. For example, the demographic transition in rural Mexico shrinks the settled workforce for reasons that have nothing to do with U.S. policy, or farmers who never hired settled workers started hiring H-2A workers simply because the program became easier to use. Separating substitution from these alternatives requires more than the aggregate pattern, and it is worth doing, because the rate at which one workforce replaces the other is a number the policy debate needs.

The answer matters for three reasons. First, it informs policy aimed at assuring adequate farm labor: whether the H-2A program can supply agricultural labor when the available settled workforce contracts. Whether the program absorbs contractions in the settled workforce driven by immigration enforcement, and at what rate, has not been established. Second, H-2A workers are considerably more expensive than settled non-citizen labor, because employers must pay them a wage floor above the local market rate and provide housing and transportation, which settled workers do not require. A substitution rate near one therefore implies that the same labor is obtained through a costlier channel, and a rate above one implies higher labor input per unit of output.[1] Third, the rate of substitution shapes how

[1]The wage floor has been rewritten and litigated twice since 2025, after the sample period ends (Congressional Research Service, 2025). When the Department of Labor issued the October 2025 rule, it projected that the lower wage floor would bring about 119,000 additional H-2A workers (U.S. Department of Labor,

farmers adjust to the loss of settled workers. The program's fixed costs make it difficult for small operators to benefit from it, so enforcement may lead to farm consolidation or to more mechanization.

To find the substitution rate, I use a panel of 740 commuting zones from 2008 to 2024. The regressor of interest is the settled non-citizen share of agricultural laborers. I instrument that share with each zone's settled non-citizen share in 2000 interacted with two national enforcement series, the real Border Patrol budget and border encounters during the agricultural season. Enforcement removes settled workers, and it removes more of them where there were more to remove.

There is a vast literature on enforcement policies, H-2A, and impacts on agriculture. Castillo and Charlton (2023) is the closest study that explores what drives H-2A visa demand. The study shows that the housing boom of the 2000s pulled workers out of agriculture, which led to higher H-2A employment within commuting zones. Arteaga and Shenoy (2022) find that a decline in Mexican migration raises wages and H-2A certifications. Castillo et al. (2023) attribute the growth to declining domestic farm employment; their estimates show eighty new H-2A positions certified for every hundred domestic employees lost, over a sample ending in 2017.

This paper differs from past studies by using enforcement policy rather than labor supply changes, a margin policymakers can control. The workforce is also defined differently. Castillo et al. (2023) measure the decline among citizen farm workers while I measure it among settled non-citizens, the group most affected by enforcement, since a majority of them lack work authorization. The distinction matters for what the estimate means. In this paper's setup, citizen farm employment rose rather than fell in the places that lost settled workers, so the substitution documented here is between H-2A and settled non-citizen workers rather than between foreign and domestic workers. Extending the sample through 2024 also covers the years in which most of the program's growth occurred. The closest study in question is Amuedo-Dorantes et al. (2021), who find that interior enforcement raises firms' demand for H-2B non-agricultural guestworkers. They regress H-2B requests on a county index of local enforcement policies, whereas I instrument the settled farm workforce with national enforcement series interacted with baseline exposure.

Previous studies have also established that enforcement reduces the supply of immigrant farm labor. The effects of interior enforcement programs like 287(g) agreements and E-Verify mandates have been shown to reduce immigrant farm labor and raise farm costs (Kostandini et al., 2014; Charlton and Kostandini, 2021; Ifft and Jodlowski, 2022; Luo and Kostandini, 2022). Immigration arrests reduce non-citizen farm labor supply immediately (Gutiérrez-Li

Employment and Training Administration, 2025), a forecast about the margin this paper estimates.

and Rubalcaba, 2025), and the effect operates partly through a chilling effect rather than through removals alone, as fear of raids and audits leads some workers to reduce hours or temporarily leave the workforce. Other broader studies document the labor market effects of enforcement on immigrant populations (East et al., 2023; East and Velásquez, 2024).

U.S. farms have experienced shortages of farm labor before. When the Bracero program ended in 1964, domestic workers did not replace the excluded Mexican workers; instead, farms mechanized or changed crops (Clemens et al., 2018). Domestic applicants for farm jobs are nearly nonexistent (Clemens, 2022), and farms are seen to adjust on other margins, through mechanization, crop choice, and imports (Rutledge and Mérel, 2023; Hamilton et al., 2022; Charlton, 2025). In method, the paper follows the shift-share literature (Goldsmith-Pinkham et al., 2020; Borusyak et al., 2025), with enforcement series as the shifters. Identification rests on the exposure share.

The main results are as follows. Farms hire about one H-2A worker for every settled non-citizen worker lost, which amounts to 0.7 work-years of guestworker labor per settled work-year lost once contract lengths are accounted for. Substitution accounts for roughly 29 percent of the program's growth over the period. A series of falsification tests supports the causal reading: the response appears only where the program can be used, follows farm-labor exposure rather than immigrant density or geography, and does not run through domestic workers.

The rest of the paper proceeds as follows. Section 2 describes the program, the workforce, and the data. Section 3 sets out the specification, why OLS is expected to fail, and the instruments. Section 4 reports the main estimates, the event studies, the aggregate attribution, and inference for the substitution rate. Section 5 tests the identifying assumption one threat at a time. Section 6 concludes.

2 Background and Data

2.1 The H-2A program

The Immigration Reform and Control Act (IRCA) of 1986 created the H-2A visa program and also made it unlawful to knowingly hire unauthorized workers on farms. The program allows agricultural employers to hire foreign farm workers for temporary or seasonal jobs, meaning jobs that last less than a year. Year-round operations such as dairies do not qualify. To hire through this program, an employer must demonstrate a failed attempt to recruit domestic workers. They are required to pay the Adverse Effect Wage Rate, a regionally set wage floor above the local market rate, provide housing that passes inspection, and cover

transportation to and from the worker's home country.

These requirements raise costs and require delays to secure workers. The high fixed cost of participation discourages H-2A use by small operations that need only a few seasonal workers. Program use is consequently concentrated among large farms and among farm labor contractors, who recruit H-2A workers and supply them to multiple growers (Castillo et al., 2024; Rutledge et al., 2026). Two consequences matter for the analysis. Because entry is slow and costly, a farm that loses workers cannot replace them through the program quickly, particularly if it has never used H-2A before. And because the program excludes year-round work, only farms whose jobs are seasonal can replace them at all.

Workers qualify for the program through an employer, not on their own. The employer must first obtain a temporary labor certification from the Department of Labor, showing that too few U.S. workers are available and that the hiring will not adversely affect the wages of similarly employed U.S. workers, and then petition U.S. Citizenship and Immigration Services for the workers, who apply for visas abroad. Petitions are generally approvable only for nationals of countries on a list the Department of Homeland Security publishes each year.[2] Workers must also be admissible under immigration law: unlawful presence in the United States of more than 180 days triggers a three-year bar on reentry, more than one year a ten-year bar, and a prior removal order its own bar.[3] Settled workers removed by enforcement are therefore generally ineligible to return through the program for years afterward. The workers H-2A supplies are different people from the workers enforcement removes.

Adoption was minimal for two decades after the program's creation and then accelerated over the sample period, as Figure 1 shows. Growth has occurred in all states, and it is largest in the fruit, vegetable, and nursery regions of California, the Pacific Northwest, the Southeast, and Florida.

2.2 The settled non-citizen workforce

After the Bracero program ended in 1964, U.S. agriculture came to rely on unauthorized migration from Mexico, sustained by migration networks and limited enforcement (Massey et al., 2016; Hanson, 2006). Many of these migrants established residence in the United States, returned to the same regions and often the same employers each season, and became the core of the seasonal workforce in labor-intensive crops. This workforce is now shrinking for two reasons. The first is demographic: rural Mexico has been transitioning out of farm work for two decades, and the supply of new migrants has fallen with it (Fan et al., 2015; Hertz and

[2]USCIS, "H-2A Temporary Agricultural Workers"; 8 CFR 214.2(h). Mexico has been on the eligible-country list in every year of the sample period.

[3]INA §212(a)(9).

Zahniser, 2013; Charlton et al., 2019). The second is policy: interior and border enforcement intensified over the sample period, and a substantial literature establishes that enforcement reduces immigrant farm labor supply at the local level (Kostandini et al., 2014; Charlton and Kostandini, 2021; Ifft and Jodlowski, 2022; Luo and Kostandini, 2022; Gutiérrez-Li and Rubalcaba, 2025). Because the demographic decline is a national trend that also raises H-2A demand, the observed decline in settled workers cannot be treated as an exogenous shock; the paper requires variation that originates in enforcement, which the instruments supply.

Roughly one third of the U.S. crop workforce lacks work authorization (Passel and Krogstad, 2024; Charlton, 2025), and it is this population that enforcement reduces. Legal status, however, is not observed in any available data set with geographic detail: the National Agricultural Workers Survey (NAWS), the only U.S. survey that records self-reported status, is not identified below the region level, and residual estimates exist only nationally. The paper therefore uses settled non-citizen status as the observable proxy. In the fiscal 2021–2022 NAWS microdata, settled non-citizens are 57.7 percent of hired crop workers and 66.8 percent of them are unauthorized (U.S. Department of Labor, Employment and Training Administration, 2023), so 38.5 percent of hired crop workers are unauthorized settled non-citizens. Consequently, my measure tracks the population subject to the enforcement instruments. My measure excludes H-2A workers because it excludes workers who lived abroad in the prior year; Appendix F reports the composition in full and the diagnostics that confirm the exclusion.

The ACS likely misses some of this settled workforce: it counts fewer settled non-citizen workers than the Census of Agriculture implies (Appendix E). The effect of the undercount on the substitution rate depends on whether it is steady or moves with enforcement. Suppose the ACS misses the same fraction of settled workers every year. Then the counted decline is smaller than the actual decline: when twenty workers leave, a survey that sees half of them records a decline of ten. The denominator is too small and the rate overstates replacement. Suppose instead that workers become less likely to respond to the survey as enforcement rises. Then the counted decline is larger than the actual decline, because part of it is workers going uncounted rather than workers leaving. The denominator is too large and the rate understates replacement. This case also matters for identification: the instrument could lower the count without any worker leaving, and the design cannot separate the two. The two cases bias the rate in opposite directions.

2.3 Data and sample

The analysis panel covers 740 commuting zones, defined by the 1990 Tolbert-Sizer delineation used by Autor et al. (2013), from 2008 to 2024, excluding 2020, when the Census Bureau did not release a standard one-year American Community Survey.[4] County and PUMA data are aggregated to commuting zones with the Dorn crosswalks. The panel contains 11,840 commuting zone-year cells, of which 11,672 have a non-missing settled non-citizen share and form the estimation sample. Appendix A gives construction details and sources for each component.

The outcome is H-2A positions certified per commuting zone and year, from the Department of Labor's Office of Foreign Labor Certification disclosure files. Each record is an employer application. It reports the workers requested and certified, the requested employment dates, and the worksite location. I deduplicate applications by case number and assign them to commuting zones by worksite county. Certifications rather than arrivals are the standard outcome in this literature (Castillo and Charlton, 2023), being available for every year, geographically precise, and closely tied to employer demand. The regressor of interest is the settled non-citizen share of agricultural laborers, from the American Community Survey via IPUMS, expressed in percent. An agricultural laborer is an employed person aged 16 to 64 in an agricultural industry and a hired farm-labor occupation, a definition that excludes operators, managers, and clerical workers and isolates the hired workforce corresponding to the NAWS population. A laborer is considered a settled non-citizen if not a U.S. citizen and not living abroad one year earlier. I refer to this measure as PANCAL, for prior-arrival non-citizen agricultural labor.

The instruments interact each commuting zone's exposure to enforcement with national enforcement series. Exposure is the zone's settled non-citizen share of agricultural laborers in 2000, from the 5 percent sample of the decennial census with the same industry, occupation, and citizenship filters; years in the United States replaces the migration question, which the census does not ask. Fixing exposure eight years before the panel begins makes it predetermined. The national shifters are the log real Border Patrol budget and the log number of U.S.-Mexico border encounters during March through September, the window in which H-2A requested start dates concentrate. Crop composition controls come from the 2002 Census of Agriculture: the vegetable share of cropland, interacted with year dummies, allows zones with different baseline crop intensity to follow different H-2A paths. Regressions are weighted by year-2000 agricultural employment, so that estimates are per unit of farm

[4]For 2020 the Census Bureau released one-year ACS estimates only as experimental data, because the COVID-19 pandemic produced atypically low response rates. I exclude the year rather than combine the experimental file with the standard releases.

labor rather than per commuting zone.

Table 1 reports summary statistics for the estimation sample and for commuting zones above and below the median year-2000 exposure. High-exposure zones employ roughly four times as many H-2A workers per year as low-exposure zones, have about three times the agricultural workforce, a much higher Hispanic population share and vegetable cropland share, and lie closer to the U.S.-Mexico border. These differences motivate two features of the analysis: the crop composition controls interacted with year, and the tests of whether the estimated effect follows farm labor rather than immigrant density or border proximity.

3 Empirical Strategy

This paper asks how many H-2A workers are hired, on average, for every PANCAL worker lost. The simplest estimator to answer this question in the absence of endogeneity is ordinary least squares (OLS). This section explains why OLS is expected to fail, sets out the strategy used instead, and explains why that strategy identifies the causal effect.

3.1 Specification

The estimating equation is

$$\text{H2A}_{it} = \delta \, \text{PANCAL}_{it} + \mathbf{X}_{it}'\beta + \gamma_i + \lambda_t + \varepsilon_{it}, \tag{1}$$

where i indexes the 740 commuting zones and t the years 2008 to 2024, excluding 2020. H2A_{it} is the number of H-2A positions certified in commuting zone i in year t, and PANCAL_{it} is the settled non-citizen share of agricultural laborers, in percent. Both variables enter in levels, not logs. γ_i and λ_t are commuting zone and year fixed effects, respectively. $\mathbf{X}_{it}$ is the 2002 vegetable share of cropland interacted with year dummies, which allows zones with more vegetable acreage to follow their own path. The regression is weighted by year-2000 agricultural employment, so that δ measures the effect per unit of farm labor rather than per place, and standard errors are clustered by commuting zone.

The parameter of interest is δ, which captures the change in the number of H-2A workers hired for a one percentage-point change in the PANCAL share. The sign is expected to be negative if farms substitute H-2A workers for settled non-citizen agricultural workers. For ease of interpretation, δ is later converted into headcount and full-time-equivalent substitution rates.

3.2 Why ordinary least squares is expected to fail

There are several reasons why the OLS estimate of equation (1) is biased, and the bias can be toward or away from zero. The commuting zone fixed effects account for fixed differences across commuting zones, such as farm size, crop suitability, and distance to the border. The year fixed effects absorb national movements over time like federal wage rules and commodity prices. What remains is local variation that moves over time and affects both the settled share and H-2A hiring. I discuss these channels below.

The first is omitted variable bias from local labor demand shocks. The crop controls in $\mathbf{X}_{it}$ absorb the part of labor demand that follows from baseline crop composition, but labor demand also moves within that composition, with prices, weather, and local expansion, for example when growers add acreage or a packing plant opens nearby. The error component can be written as $\varepsilon_{it} = \theta D_{it} + \nu_{it}$, where D_{it} is the local labor demand shock that remains after the controls. A positive demand shock raises H-2A hiring directly ($\theta > 0$) and also attracts or retains settled workers, so $\operatorname{cov}(\text{PANCAL}_{it}, D_{it}) > 0$. The omitted-variable formula then gives

$$\operatorname{plim} \hat{\delta}^{OLS} = \delta + \theta \, \frac{\operatorname{cov}(\text{PANCAL}_{it}, D_{it})}{\operatorname{var}(\text{PANCAL}_{it})} > \delta,$$

so with $\delta < 0$ the OLS estimate is pushed toward zero and can even turn positive. Places that are growing hire more of both types of workers.

The second type of endogeneity comes from measurement error in the regressor. Classical measurement error attenuates the coefficient toward zero. The PANCAL share is constructed from thin ACS cells in most commuting zones, so much of its year-to-year variation within a zone is sampling noise. An instrument uncorrelated with the sampling noise corrects this attenuation as well as the bias from omitted variables.

Two other potential sources of bias run in the opposite direction, making δ more negative. The first is reverse causality: if the rise in H-2A hiring is what pushes PANCAL workers out, which could happen because farmers gain access to these workers through a legal channel, there will be a substitution rate, but the causal direction would be reversed. The second runs through the share's denominator: H-2A workers who appear in the ACS are counted among agricultural laborers but not among settled non-citizens, so the share can fall because the denominator grows, without any settled worker leaving. Both would push the OLS estimate away from zero.

3.3 Instruments

I instrument the settled share with two shift-share instruments that interact a fixed measure of each commuting zone's exposure to enforcement with a national enforcement series. I construct the Bartik instruments as follows:

$$(2) \qquad B_{it}^{BP} = s_{i,2000} \times \ln(\text{BP budget}_t), \qquad B_{it}^{enc} = s_{i,2000} \times \ln(\text{ag-season encounters}_t),$$

where $s_{i,2000}$ is the exposure share, the baseline PANCAL share of agricultural laborers in commuting zone i in 2000. It is constructed from the decennial census, eight years before the panel begins. The two shift components are national series that capture enforcement pressure each year: the real Border Patrol budget and the number of agricultural-season border encounters.

The logic behind these instruments is that if enforcement removes settled non-citizen workers, the effect should be larger in places that depend more on these workers. A zone with a high 2000 share should see a larger decline in its PANCAL share when the budget rises or border activity intensifies than a zone with a low share. The instrument predicts the decline from baseline exposure and national conditions alone, so it is unaffected by the zone's own H-2A demand.

The two shifters differ in character. The budget grows monotonically and captures the long-run expansion of enforcement capacity. On the other hand, the encounter series is non-monotonic and captures border activity in the months when farms hire. The budget instrument is the benchmark throughout the paper because its exogeneity argument is the simplest: federal appropriations are set nationally and unaffected by any commuting zone's farm labor market. The two instruments use largely distinct identifying variation; under homogeneous effects, they identify the same parameter.

3.4 Identifying assumption

Each instrument interacts one exposure share with one national time series; the overidentified specification uses both. Identification comes from the baseline exposure shares, following the exogenous-shares framework of Goldsmith-Pinkham et al. (2020). The shifters put higher weight on years with more enforcement pressure, but they do not supply independent variation. The identifying assumption is therefore that, conditional on the fixed effects and the crop controls, commuting zones with higher year-2000 exposure would have followed the same H-2A path as less exposed zones had enforcement not intensified. Equivalently, the instrument affects H-2A certifications only through the contemporaneous settled share.

Three features of the instrument design support this assumption. First, the exposure share predates the H-2A expansion as well as the enforcement buildup. I use the year 2000 for the exposure share because it allows me to use the decennial census, which covers a 5 percent sample, and because it predates movement in both the outcome and the settled share, so the variation can be argued to be exogenous. Second, the exposure share is specific to settled non-citizen agricultural labor rather than the general immigrant population or domestic agricultural workers. This enables me to focus on the specific population expected to move due to enforcement and the one the H-2A program can replace. Immigrant density outside agriculture and proximity to the border do not produce the same effect, and the effect is absent in sectors where H-2A is unavailable. Third, the event studies show that low and high baseline PANCAL exposure commuting zones are on similar paths before enforcement kicks in, and that the H-2A program responds to the decline in the PANCAL share rather than preceding it. The one pre-trend that does exist, in domestic employment, does not affect the estimate once controlled for.

3.5 First stage, reduced form, and event studies

The first stage regresses the PANCAL share on the instruments with the same controls and fixed effects,

$$\text{PANCAL}_{it} = \pi^{BP} B_{it}^{BP} + \pi^{enc} B_{it}^{enc} + \mathbf{X}_{it}'\beta + \gamma_i + \lambda_t + u_{it}, \tag{3}$$

and the reduced form regresses H-2A certifications on the instruments directly,

$$\text{H2A}_{it} = \rho^{BP} B_{it}^{BP} + \rho^{enc} B_{it}^{enc} + \mathbf{X}_{it}'\beta + \gamma_i + \lambda_t + e_{it}. \tag{4}$$

The second stage is equation (1) with the PANCAL share replaced by its fitted value from the first stage. $\hat{\delta} = \hat{\rho}/\hat{\pi}$ for just-identified regressions. Table 2 reports all three stages, the Kleibergen-Paap F-statistic for instrument strength, and the Anderson-Rubin test, which remains valid when the first stage is weak. Under homogeneous effects, the Hansen J test checks whether the two instruments identify the same underlying parameter.

To trace the dynamic response, I replace the enforcement shift component with a full set of year dummies:

$$Y_{it} = \sum_{k \neq 2008} \theta_k \left(s_{i,2000} \times \mathbf{1}[t = k] \right) + \mathbf{X}_{it}'\beta + \gamma_i + \lambda_t + \varepsilon_{it}. \tag{5}$$

For the first-stage event study, Y_{it} is PANCAL share. For the reduced-form event study, Y_{it} is

H-2A certifications. With 2008 as the reference year, each θ_k measures how the gap between more and less exposed commuting zones differs relative to the same gap in 2008.

4 Results

4.1 Graphical evidence

Figure 2 plots the identifying variation directly. Panel (a) shows that commuting zones with a higher baseline PANCAL share saw larger declines in contemporaneous PANCAL between 2008 and 2024, with a slope of −0.18 (SE 0.04). This supports the relevance condition: places that historically had a larger share of settled workers are the ones where the contemporaneous PANCAL decline was largest. Panel (b) shows that those same commuting zones also experienced larger increases in H-2A worker certifications over the same window, with a slope of 76.4 (SE 18.1).

Figure 3 shows how the two variables look spatially across the country. Panel (a) shades commuting zones by baseline PANCAL exposure, and panel (b) shades them by the change in H-2A workers between the 2008-2010 and 2022-2024 averages. The darkest zones in panel (b) overlap closely with the darkest baseline PANCAL exposure zones in panel (a), especially in California, Florida, and the Yakima Valley and Columbia Basin of Washington. The program's growth is highly concentrated such that the thirty zones in the top class account for two thirds of the national increase.

4.2 Main estimates

Table 2, column (1), gives the OLS coefficient $\hat{\delta}^{OLS} = -2.11$ (SE 1.61), which is economically small and statistically insignificant. The OLS coefficient is subject both to classical measurement error from the thin-cell ACS share measure and to endogeneity. Measurement error attenuates the coefficient mechanically and Table 3 quantifies the extent of each bias.

Columns (2) to (4) report the two-stage least squares estimates. Instrumenting the PANCAL share with the Border Patrol budget Bartik yields $\hat{\delta} = -156.7$ (61.3); the agricultural-season encounter Bartik yields −226.5 (65.7); the overidentified specification, which uses both, yields −178.9 (60.7). The coefficients are interpreted as change in number of H-2A workers per commuting zone for a one percentage-point change in the PANCAL share, so the negative sign means that a decline in the settled share raises H-2A hiring. For ease of interpretation I scale each coefficient by the workforce of the commuting zones that identify it. Two-stage least squares averages zone-level responses with weights that depend on each zone's baseline exposure and on how strongly the instrument moves its share, the Rotemberg weights of

Goldsmith-Pinkham et al. (2020). Large, highly exposed zones dominate: California alone carries about half of the weight, and a few zones whose shares move against the national pattern receive small negative weights. The weighted average workforce is $\bar{N}^* = 15{,}426$ workers under the benchmark instrument, so a one percentage-point decline in the share represents about 154 workers, and the estimates imply a substitution rate of 1.02 on the benchmark instrument and between 1.0 and 1.6 across the three specifications. H-2A contracts, however, are not year-round. The average contract lasts 5.7 months (Appendix Figure B1), while settled workers average roughly eight months of farm work per year.[5] Accounting for this difference, the full-time-equivalent substitution rate is 0.72 on the benchmark instrument and between 0.7 and 1.2 across specifications. Farms in the zones that identify the estimate replace lost settled workers approximately one for one in heads, and replace about three quarters of the lost labor time.

Panel B shows that the first stage is negative and precise for both instruments. The Kleibergen-Paap (KP) F-statistics test whether the instruments move the settled share strongly enough for standard inference; they range from 14.8 to 32.5 and lie above the 15 percent Stock-Yogo critical values. The first stage has the expected sign: when enforcement intensifies nationally, the PANCAL share falls by more in commuting zones that had more settled workers to begin with. The Anderson-Rubin p-values show that inference does not rest on first-stage strength. Panel C reports the reduced-form coefficients, which show that when enforcement intensifies, H-2A hiring rises by more in the more exposed zones. Each reduced-form coefficient equals the first stage times the second stage up to rounding ($55.8 \approx 0.356 \times 156.7$). When both instruments enter jointly, each remains individually significant, so the two enforcement series carry independent identifying variation. The Hansen J test does not reject the overidentifying restriction ($p = 0.167$), consistent with the two instruments identifying the same underlying parameter.

4.3 The gap between OLS and IV

The IV estimates are nearly 75-110 times larger than OLS. To evaluate this gap, I decompose the OLS-IV gap using an errors-in-variables (EIV) correction based on a direct estimate of the reliability of the PANCAL share. Using a Monte Carlo framework, I estimate the reliability ratio by holding fixed each commuting zone's true agricultural-laborer composition across pooled years. For each commuting zone-year cell, I redraw the ACS sample using its actual respondent count, recalculate the share, and residualize it under the main specification.

[5] U.S. Department of Labor, National Agricultural Workers Survey, Public Data Tables, Table 2: hired crop workers averaged 35 weeks of farm work per year across the six survey cycles spanning 2007-2020 (35, 34, 35, 33, 35, and 39 weeks).

Comparing the resulting pure sampling variance with the total residual variance yields an estimated reliability of $\hat{R} = 0.180$. This implies that less than 20 percent of the within-zone variation in the measured share is true signal after fixed effects and controls are added. Table 3 reports the decomposition. Dividing the OLS estimate by $\hat{R}$ produces an EIV-corrected coefficient of -11.7. A commuting-zone block bootstrap that re-estimates both the coefficient and $\hat{R}$ gives a 95 percent confidence interval of $[-28.1,\ 8.2]$. Even under a lower reliability of $R = 0.10$, the corrected OLS estimate reaches only -21.1. Measurement error therefore accounts for 6.2 percent of the gap between the OLS estimate (-2.1) and the benchmark IV estimate (-156.7); the remaining 93.8 percent reflects endogeneity. This is consistent with the omitted-variables argument in Section 3: positive local labor demand shocks raise H-2A hiring while attracting or retaining settled workers, inducing a positive correlation between the error term and the PANCAL share and biasing OLS toward zero. The instrument therefore corrects endogeneity bias rather than merely adjusting for sampling noise.

4.4 Event study dynamics

The main estimates report a single coefficient for a pattern spanning sixteen years. Figure 4 shows how the relationship between baseline exposure and each variable, the settled share and H-2A hiring, evolves over time. I interact the year-2000 PANCAL share with a set of year dummies, with year 2008 as reference. Each coefficient captures how the gap in each variable evolves in high PANCAL versus low PANCAL exposure regions relative to year 2008.

Panel (a) shows the first stage. The coefficients are close to zero through 2014 and are not jointly different from zero over 2009-2012. They become significant at the 5 percent level from 2015. This means that exposed commuting zones lost settled non-citizen workers faster than unexposed ones from 2015 onward, and the gap widened further after 2019. Panel (b) shows the reduced form. The early coefficients are small and negative, and jointly different from zero. The coefficients become positive and significant at the 5 percent level from 2019, rise through 2023, and remain large in 2024. Two features are important to note. First, if exposed commuting zones had been hiring more H-2A workers than other zones before enforcement intensified, the early coefficients would be positive. Instead they are negative, which works against the effect rather than inflating it. Second, the coefficients through 2015 are small relative to the later period, about 8 percent of the 2023 peak at most, and largest in 2009, the recession year.

The two panels together depict the substitution pattern within a timeline. The settled workforce in exposed commuting zones begins to decline in 2015, and H-2A hiring in the same zones rises significantly from 2019. The gap between the two dates describes the timing;

it is not an estimated lag, and Appendix C shows that the design cannot identify the lag length directly. The reduced form also accelerates in 2019, which coincides with the larger decline in the settled worker share.

4.5 Aggregate attribution

Table 4 shows how much of the growth in H-2A employment can be explained by the PANCAL decline. Between the 2008–2010 and 2022–2024 averages, summing over the 735 commuting zones observed in both windows, H-2A employment increased by 298,923 workers, while the settled non-citizen agricultural workforce fell by 83,912. Multiplying the decline of 83,912 workers by the benchmark substitution rate gives a predicted increase of $1.02 \times 83{,}912 = 85{,}243$ H-2A workers, which accounts for 29 percent of the observed growth. For the encounter and overidentified rates the shares are 46 and 34 percent, respectively. The settled-worker decline is therefore a large contributor to the program's growth. The remaining contributors include places that adopted H-2A without a history of reliance on settled non-citizen workers; Figure 3 shows moderate growth across the Midwest, where fruit and vegetable production is limited and H-2A use was minimal at baseline. They also include places where both the settled workforce and H-2A employment rose because local labor demand grew, variation the instrument deliberately sets aside.

4.6 Interpreting the magnitude

The main coefficients in Table 2 report cluster-robust standard errors whose confidence intervals assume that $\hat{\delta}$ is approximately normal in large samples. With a sample of 740 commuting zones, this assumption would be reasonable, but when a large share of the weight is carried by a few commuting zones, the effective sample can be far below 740. I therefore run a commuting-zone block bootstrap, which does not rely on this approximation. In each of the 999 replications, I draw the 740 commuting zones with replacement, re-estimate two-stage least squares, and form the substitution rate $|\hat{\delta}|/(0.01 \times \bar{N}^*)$. Table 5 reports the resulting percentile intervals. The denominator is held at the full-sample first-stage average, $\bar{N}^* = 15{,}426$, so the rate's uncertainty comes from $\hat{\delta}$. Appendix D reports a version that also recomputes $\bar{N}^*$ in each replication; its median is also near one, and its tails are driven by weakly identified replications. Figure 6 shows that the sampling distribution is right skewed, which is why the intervals are percentile based rather than symmetric.

Three conclusions follow from the intervals. First, the headcount rate is centered near one: the point estimate is 1.02, the bootstrap median is 1.11, and the probability that the rate falls below one is 42 percent, so the data do not separate replacement below one for one from

replacement above it. Second, replacement well above one for one cannot be ruled out; the upper bound of the featured interval is 3.2. Third, the labor-time rate is centered just below one, with a median of 0.79 and an interval of $[0.20, 2.26]$, so exact one-for-one replacement of labor time cannot be rejected. A delta-method standard error that assumes normality is 0.40, well below the bootstrap standard error of 1.04, and would overstate precision considerably.

The right tail has one identifiable source, documented in Appendix D. Farm labor is geographically concentrated, and bootstrap draws that underweight the California core thin out the identifying variation, weaken the first stage, and inflate $|\hat{\delta}|$. Conditioning on replications with a first-stage F above 10 trims the upper bound from 3.2 to 2.0 and moves the probability of a rate below one from 42 to 53 percent, so weak first stages inflate the rate rather than shrink it. Appendix E reports the rate under alternative assumptions about ACS coverage of the settled workforce.

5 Robustness

The causal claim of the main results holds if the identifying assumption is valid: the Bartik instrument, baseline PANCAL exposure times national enforcement, affects the outcome variable, H-2A certifications, only through changes in the main regressor, the contemporaneous PANCAL share. There is no single way of testing this, so in this section I run a number of tests, each checking one way in which the estimate could be wrong. The same tests also justify several choices made in constructing the variables and the specification: why the main regressor is the settled non-citizen share rather than a broader immigrant measure, why the crop controls are constructed the way they are, how the estimate changes when the definition of the agricultural workforce is widened or narrowed, and why two instruments are used and how they differ.

5.1 Testing the exposure measure against immigrant density and border proximity

I test whether the substitution rate proposed above picks up the decline of settled non-citizen agricultural workers or simply captures regions with higher immigrant density in general. The two are hard to tell apart. Suppose two commuting zones have the same baseline PANCAL share. The first has the majority of its immigrant workers in agriculture. In the second, agricultural workers are a small fraction of a much larger immigrant population working in non-agricultural sectors. Enforcement affects both zones' farm workers similarly, but in the second it also affects the larger immigrant workforce outside agriculture. If the second kind

of place also expanded H-2A hiring when enforcement tightened, for reasons unrelated to losing farm workers, the instrument would be picking up immigrant density in general and the coefficient would overstate substitution.

To test the above, I build a non-agricultural version of the Bartik instrument in which the exposure share is the settled non-citizen share of the non-agricultural workforce, interacted with the same enforcement shifters. To make the non-agricultural Bartik comparable to the agricultural one, I restrict the non-agricultural population to low-education workers. I then enter both Bartiks in one reduced form,

$$\text{H2A}_{it} = \pi_{ag} B_{it}^{ag} + \pi_{nonag} B_{it}^{nonag} + \mathbf{X}_{it}'\beta + \gamma_i + \lambda_t + \varepsilon_{it}, \tag{6}$$

where B_{it}^{ag} is the agricultural Bartik of the main specification and $B_{it}^{nonag} = s_{i,2000}^{nonag} \times \ln(\text{shift}_t)$ is its non-agricultural counterpart. If immigrant density were what mattered, the two would split the effect or the non-agricultural Bartik would dominate. Table 6 shows the opposite. The agricultural Bartik keeps a positive, significant coefficient, while the non-agricultural Bartik has the wrong sign and is imprecise. I also add the contemporaneous non-agricultural settled non-citizen share as a control in the main specification, and the estimate moves modestly, and away from zero, as Panel B of Table 6 shows.

Enforcement studies have used distance from the Mexican border as a variable that measures enforcement exposure. Being closer to the border would imply higher unauthorized activity and hence higher enforcement exposure. To see whether the variation actually comes from activity closer to the border, I interact negative log distance to the border with the same budget shifter as in the main specification. The second column of Table 6 shows that this instrument is weak (first-stage F of 4.7) and gives an imprecise estimate. When put in a reduced-form horse race with the share-based Bartik, the share-based Bartik survives. Thus, proximity to the border does not identify the effect once the location of settled non-citizen workers is accounted for.

One exclusion story is specific to farms. Growers in high-exposure zones could see enforcement rising and enter the H-2A program before losing any settled workers, as insurance. H-2A hiring would then rise while the PANCAL share was unchanged, and the non-agricultural comparison above would not detect it. The event studies do. Anticipation of this kind would place the H-2A response ahead of the PANCAL decline. The order in Figure 4 is the reverse. The first stage moves in 2015 and the reduced form in 2019, and the H-2A coefficients show no positive pre-trend.

5.2 Testing for a response where H-2A is unavailable

H-2A is a seasonal program, limited to jobs that last less than a year. There are sectors that employ settled non-citizen workers but may not have access to H-2A because they require year-round employment. Livestock production is such a sector: dairy, its largest employer, along with cattle, hog, and poultry operations, relies on year-round labor, so enforcement may lead to a decline in settled workers but an H-2A alternative does not exist in most cases. If an H-2A response exists where the visa is not available, that would mean my results are picking up a different channel. If the response is substitution, it should be absent in livestock.

The main instrument's exposure is a broad definition of all agricultural laborers. Here, I split it into its two sectors. I define the crop share as the settled non-citizen share of laborers in crop production and agricultural services (IPUMS 1990 industries 10 and 20) and the livestock share as the settled non-citizen share of laborers in livestock production (industry 11), for each commuting zone and year.

I ask two questions of the livestock Bartik. First, does enforcement actually remove livestock settled workers, so that the placebo carries a real shock? The contemporaneous livestock share is the dependent variable in the first stage. Second, does that shock raise H-2A hiring? The second question is answered by letting the two components compete in one reduced form,

$$\text{H2A}_{it} = \pi_{crop} B_{it}^{crop} + \pi_{lvstk} B_{it}^{lvstk} + \mathbf{X}_{it}'\beta + \gamma_i + \lambda_t + \varepsilon_{it}. \tag{7}$$

The 2000 values of each share are interacted with the enforcement shifter to give a crop Bartik B_{it}^{crop} and a livestock Bartik B_{it}^{lvstk}, constructed exactly as the main instrument. Table 7 reports both. The first stage for livestock is negative and significant with either enforcement series, implying that enforcement drains the livestock settled workforce as it does the crop workforce and the broader agricultural workforce in the main specification. However, in the horse race, H-2A certifications respond to the crop Bartik while the coefficient on the livestock Bartik is negative and imprecise. This means that the substitution response appears only where H-2A is available.

5.3 Testing for pre-existing trends in exposed places

The substitution rate attributes the rise in H-2A hiring to the loss of settled non-citizen workers. A competing explanation attributes it to a different loss. Castillo et al. (2023) argue that H-2A growth followed a decline in domestic farm employment, and estimate that roughly eighty H-2A jobs were certified for every hundred domestic employees lost. If domestic

workers in exposed commuting zones were leaving agriculture, aging out or moving into construction and services, and H-2A replaced them, then my instrument could be crediting the settled-worker channel for a channel that runs through citizens instead. The two stories have different policy implications: one says enforcement created the vacancies, the other says demographics did.

The two explanations make different predictions for domestic employment. If H-2A were replacing departing citizens, domestic employment in exposed zones should fall as enforcement tightens. If H-2A were replacing settled non-citizens, domestic employment need not fall. I run the main specification with domestic agricultural employment, meaning U.S.-born and naturalized citizen workers, in place of H-2A workers as the outcome. Panel A of Table 8 shows that domestic employment does not fall; it rises, and by a lot. Taken at face value, each settled worker lost would be matched by 1.3 to 2.5 additional domestic workers, depending on the instrument. Whatever H-2A replaced in exposed zones, it was not departing citizens. The workforce that declined is the settled non-citizen one, and citizen employment in the same places grew.

That growth raises a second question. A rising domestic workforce in exposed places suggests those places were changing in ways unrelated to enforcement, most plausibly expanding, and expansion could raise every kind of hiring, H-2A included. Figure 5 confirms the suspicion. The event study of domestic employment on baseline exposure shows a gap that opens in the very first year after the reference year and grows smoothly through the period, and the joint test that the 2009-2012 coefficients are zero rejects at the 1 percent level. This is a pre-existing trend, not a response to enforcement, and it is exactly the pattern the same test did not find for H-2A in Section 4.4. It is also not an artifact of settled workers naturalizing and being reclassified as citizens: restricting to U.S.-born workers, a group no one can join by naturalizing, gives the same pre-trend, also rejecting at the 1 percent level, and most of the coefficient.

The question that matters for the substitution rate is whether it rests on this trend. I control for it directly. Let G_i be commuting zone i's growth in domestic agricultural employment over 2008-2012, which is predetermined and only weakly correlated with exposure, and add it to the main specification interacted with year dummies,

$$\text{H2A}_{it} = \delta \, \text{PANCAL}_{it} + \sum_t \theta_t \, G_i \times \mathbf{1}[\text{year} = t] + \mathbf{X}'_{it}\beta + \gamma_i + \lambda_t + \varepsilon_{it}, \tag{8}$$

so that each commuting zone can follow its own time path scaled by its pre-trend, and δ is identified only from variation orthogonal to pre-existing domestic trajectories. Panel B of Table 8 shows that the estimate does not change qualitatively. The benchmark coefficient stays

within half a standard error of its original value, and the encounter and overidentified estimates move toward it so that the two instruments agree more closely than before. The substitution estimate does not live on the domestic trend. A more aggressive control, contemporaneous domestic employment itself, is endogenous to the same shocks and is not a valid specification.

5.4 Testing whether the instruments identify the same parameter

An instrument is only as good as the argument behind it. Section 4 reported two instruments with the same baseline exposure share but different national enforcement shifters. The Border Patrol budget captures structural enforcement capacity which grows monotonically year by year while the agricultural-season encounters captures contemporaneous border movement and is non-monotonic. The two together capture different variations of the same underlying parameter (Hansen J test does not reject, $p = 0.17$). The agreement is informative because the two instruments share little of the variation the regression uses: after fixed effects and controls, the correlation of their residual variation is only 0.28. Identification comes from the exposure share and not the shifts. The comparison of instruments in this section tests whether the timing of the H-2A response follows enforcement.

Table 9 pushes on this from two directions. Panel A shows that if the shift component of the instrument is replaced with any other monotonically increasing series like consumer price index, federal health spending or a simple linear trend, it should behave like the Border Patrol budget, and it does. This is the critique of monotone shifters raised by Goldsmith-Pinkham et al. (2020). The encounter instrument is the answer to this limitation. Appendix Figure B2 plots the different national series I use here. The benefit of using the agricultural-season border encounter shift is that its variation does not resemble other monotonic series, its movements are policy driven, and yet it gives a stronger KP F-statistic. The paper treats the budget instrument as the transparent benchmark and leans on the encounter instrument, and on the overidentification test, for the claim that the variation is enforcement rather than trend.

The literature on immigrant labor supply typically uses instruments that capture national changes in a group's supply, allocated to places where that group was already concentrated (Card, 2001). To connect the estimates of this paper to this conventional design, I construct an instrument whose shift component measures the national decline in PANCAL workers, keeping the baseline PANCAL share as the exposure variable. There are two versions of this instrument: a leave-one-out Bartik, in which the national series of the PANCAL decline excludes the own commuting zone, and another that includes it. This supply-shift instrument rests on a different argument from this paper's. It allocates the entire national decline

to places by baseline shares, irrespective of the cause of that decline, rather than only its enforcement-driven part. I further run overidentification tests with the leave-one-out Bartik paired with each of the two enforcement instruments individually, and then with all three together. Panel B of Table 9 shows that the supply-shift instrument gives an estimate close to the enforcement instruments. No pairing rejects the overidentifying restriction at 5 percent. The estimate is closest to the encounter instrument's and the pairing with the budget instrument is borderline, the same gap between smooth and non-smooth shifters that Panel A documents. The substitution rate is therefore a property of losing settled workers, whatever the cause, rather than of enforcement specifically.

The tests so far show whether the instruments agree; I now turn to where their agreement comes from. Goldsmith-Pinkham et al. (2020) discuss how to interpret an overidentified IV estimate. It is a weighted average of just-identified estimates, and the weight on each measures its contribution to the first-stage covariance between the instrument and the settled share. The weights can be computed across the two instruments, across years, and across commuting zones. Table 10 reports them along all three dimensions. The budget instrument receives 68 percent of the weight and the encounter instrument 32 percent. The weighted just-identified estimates reproduce the overidentified coefficient. The weights across years are largest for 2019, 2023, and 2024, which are the surge years. The pre-2018 years carry negative aggregate weight. This implies that in those years enforcement had not yet risen and the settled share had not yet fallen relative to the panel mean, so they act as the before period that the surge years are compared against. Negative weights of this kind are a feature of the decomposition, not a sign of misbehavior (Goldsmith-Pinkham et al., 2020). Across commuting zones, California carries 49 percent of the identifying weight on the benchmark instrument and 45 percent on the overidentified estimate. That concentration is not an accident of the design. It follows from where American farm labor is and from weighting the estimate by farm employment.

5.5 Alternative specifications and samples

Table 11 reports how the controls in the main specification were chosen. Without the crop controls, the estimates are larger and less precise. Areas with higher crop demand could hire more H-2A workers even without a substantial decline in PANCAL workers, and the crop controls account for this demand. I construct three types of crop controls, each fixed at its 2002 Census of Agriculture value and interacted with year dummies: a labor-intensive crop share, which is the combined acreage of vegetables, orchards, and berries as a share of cropland; the vegetable, orchard, and tobacco shares entered as separate controls; and

the vegetable share of cropland alone. Table 11 shows that the first two absorb too much of the identifying variation and weaken the first stage. The vegetable share of cropland, on the other hand, keeps the first stage strong while absorbing the crop composition that both attracts settled workers and uses H-2A.

Section 4.6 shows that the California core anchors the substitution rate, so the natural check is to remove it. Table 12 indicates what happens to the estimates when I drop California from the sample or only the three largest zones in California. In the former case, the coefficient doubles and the first stage weakens to an F of 5.5, because most of the identifying variation leaves with the state. In the latter case, which leaves the rest of the state in, the coefficient stays within a few units of the benchmark. No rate is reported for either row. The rate divides by the workforce of the zones that identify the estimate, and without the California core that workforce falls from 14,913 to 1,065 and 3,953 workers while the first stage weakens, so the denominator is no longer stable (Appendix D). The coefficient does not depend on California; the denominator does. The unweighted specification makes the same point from the other direction. It weighs all commuting zones equally, but a percentage point decline in PANCAL in California is not comparable with the same decline in Iowa, so the coefficient is smaller in levels and the identifying workforce smaller, and the rate, 1.9, is close to the benchmark. The weighted specification measures the effect per unit of farm labor, which is what the substitution question asks. Dropping the top one percent of commuting-zone-year H-2A outcomes leaves the coefficient unchanged while the first stage weakens.

The second group varies the sample and the timing. The crop agriculture sector covers crop production and agricultural services (IPUMS 1990 industries 10 and 20); it is the sector that employs the most PANCAL and H-2A workers. Restricting the sample to crops gives a substitution rate close to two. The PANCAL construction specifically excludes recently arrived non-citizens. If they affected my estimates, including their share as a control variable would shift the main coefficient. As Table 12 shows, the coefficient barely moves, reinforcing the group-quarters evidence in Appendix F that recent arrivals do not contaminate the settled measure. Lagging the instruments by a year, which allows enforcement time to bite, gives the same coefficient but a weaker first stage. Lagging the settled share instead, which matches H-2A applications to the labor conditions observed a year earlier, gives a slightly larger coefficient, as expected from the event studies. Neither changes the estimate substantially.

The next variation widens the definition of the main regressor to all Mexican-born or all foreign-born agricultural laborers, including citizens and recent arrivals, following Castillo et al. (2023). The wider definitions give larger coefficients. These are the only two specifications in the paper where the Hansen J test rejects. This is a finding on its own. The two enforcement instruments agree when the effect is measured on settled non-citizen agricultural

labor, not on the wider definitions, where the enforcement channel is diluted. The difference comes from the inclusion of citizens, who are unaffected by enforcement, and recent arrivals, who likely include H-2A workers themselves. The settled non-citizen share is the definition on which the two instruments tell the same story, which is why it is the paper's main regressor.

Splitting the panel into two halves, 2008 to 2017 and 2018 to 2024, I re-estimate the main specification in each. As noted in Section 5.4 through the Rotemberg weights, neither half has enough identifying variation to estimate $\hat{\delta}$ precisely. This is because the enforcement variation is smaller within either window. The point estimates follow the event-study timing, small in the early half and large in the late half, so the two halves read like a before and after comparison.

Finally, Table 12, Panel B, shows what happens when the outcome variable is proportional instead of in levels. The inverse hyperbolic sine of H-2A workers as the outcome gives a coefficient of -0.089 (0.038), which means that a one percentage-point decline in the PANCAL share raises H-2A employment by about 9 percent. The log of H-2A workers, dropping the 2,531 commuting-zone-years with zero H-2A (480 of them in commuting zones that never use the program), gives a coefficient of -0.124 (0.037) with a first-stage F of 14.8, which means a 12 percent increase in H-2A employment per percentage point. The proportional transformations compress the large-H-2A zones that carry most of the identifying variation, so the implied rates, 0.59 and 0.90, are below the main specification's 1.20.

6 Conclusion

This paper estimates that farms hire approximately one H-2A guestworker for each settled non-citizen agricultural worker lost to enforcement-driven decline, replacing about three quarters of the lost labor time once contract lengths are accounted for. Substitution for lost settled workers accounts for about 29 percent of H-2A growth between 2008 and 2024. The result survives alternative national shifts, including smooth-series benchmarks and the supply-shift construction of the immigration literature, crop composition controls, and falsification tests against non-agricultural labor markets, border proximity, and sectors where the program is unavailable. Event studies show that the settled workforce in exposed places begins to decline in 2015 and that H-2A hiring responds from 2019.

The estimate is imprecise in a documented way. Farm labor is concentrated in a few places, and resampling draws that underweight them stretch the confidence interval on the headcount rate from 0.28 to 3.2. The data do not rule out replacement below one for one, and they do not rule out replacement well above it. What the estimates support is a rate centered near one.

The estimates bear on the three questions raised at the outset. First, the program does serve as the adjustment margin for enforcement-driven contractions in the settled workforce. The event studies date the settled decline from 2015 and the H-2A response from 2019; the design does not identify the adjustment speed. Second, enforcement in this period did not so much reduce the labor employed in agriculture as change who supplies it, from settled workers living in the United States to temporary workers recruited abroad on short contracts, at the higher cost the program imposes through its wage floor, housing, and transportation requirements. Third, the program is open to any employer who can meet its requirements, but its fixed costs concentrate use among large operations and labor contractors, so the replacement documented here was within reach for some employers and out of reach for others, and enforcement may be accelerating consolidation in labor-intensive agriculture.

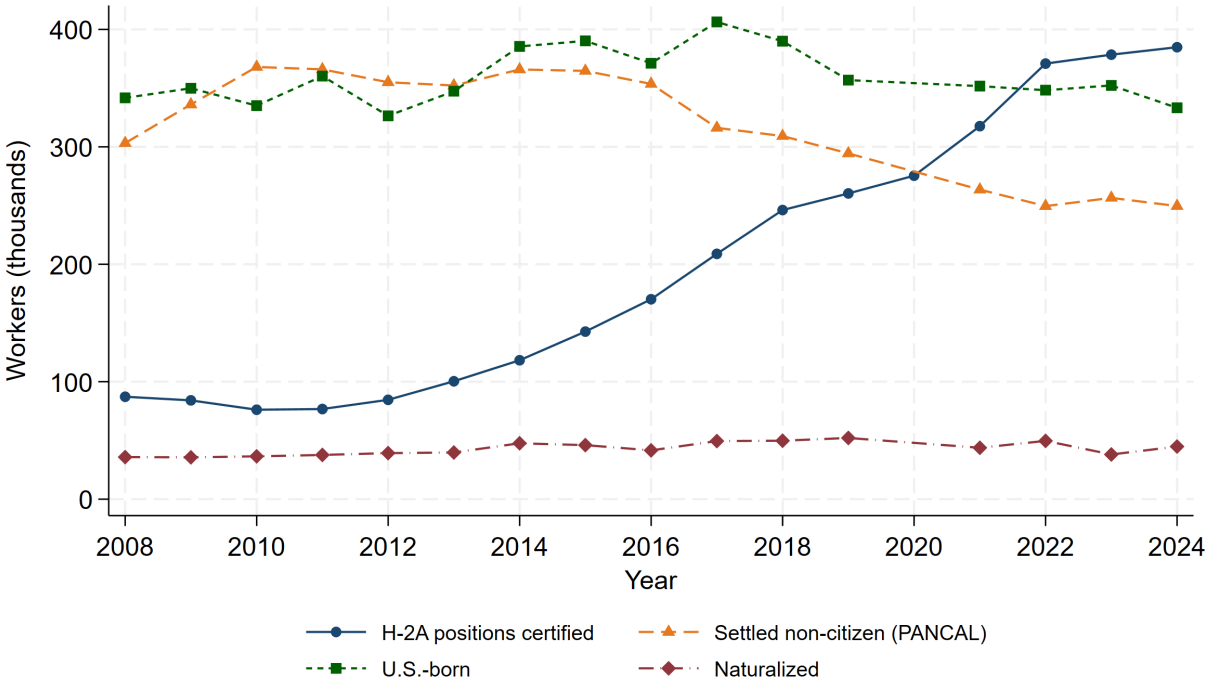


FIGURE 1. Components of the U.S. Hired Farm Workforce, 2008-2024

Notes: National totals in thousands of workers. H-2A positions certified are from the Department of Labor OFLC disclosure files, full and partial certifications, by fiscal year. The three ACS series are person-weighted counts of agricultural laborers, employed persons aged 16 to 64 in crop production, livestock production, or agricultural services in farm-worker, nursery-worker, or supervisor occupations, split by citizenship: settled non-citizen (PANCAL) workers are non-citizens who did not live abroad in the prior year, and the U.S.-born and naturalized series are citizens. The ACS did not release standard one-year estimates for 2020, so the ACS series have no value in that year. *Sources:* Department of Labor OFLC disclosure files; American Community Survey one-year files via IPUMS.

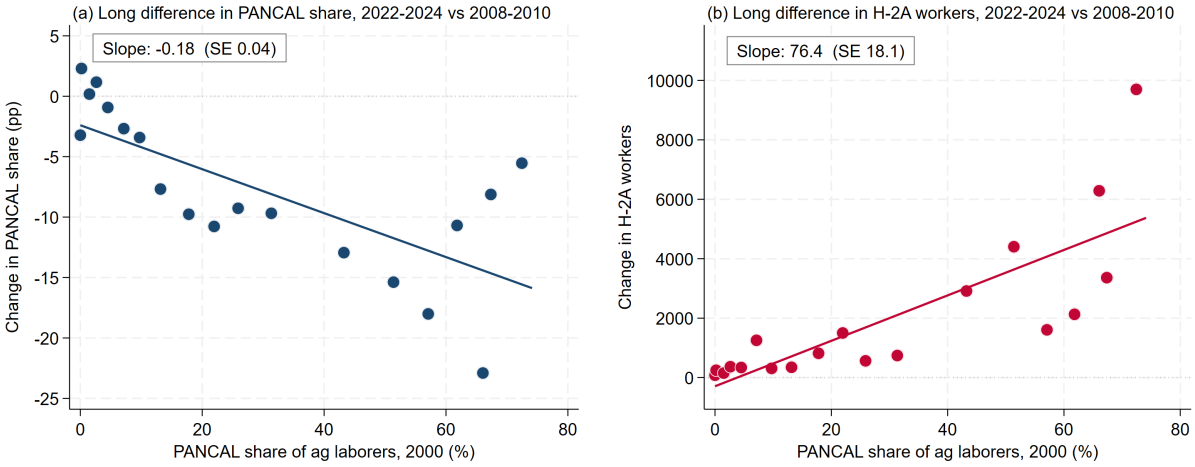


FIGURE 2. The Identifying Variation in Long Differences

Notes: Each panel plots a long difference against baseline exposure across the 740 commuting zones in the estimation sample. Baseline exposure is the PANCAL share of agricultural laborers in 2000. The long difference is the 2022–2024 average minus the 2008–2010 average, of the PANCAL share in percentage points in panel (a) and of certified H-2A workers in panel (b). Commuting zones are sorted by exposure and grouped into nineteen bins each holding approximately 5 percent of year-2000 agricultural employment; a commuting zone holding at least 5 percent by itself forms its own bin. Each dot is the employment-weighted mean of the bin. Fitted lines and the reported slopes are from regressions at the commuting-zone level, weighted by year-2000 agricultural employment, with heteroskedasticity-robust standard errors; they are not fits through the binned points. *Sources:* American Community Survey via IPUMS; Department of Labor OFLC disclosure files; Census 2000.

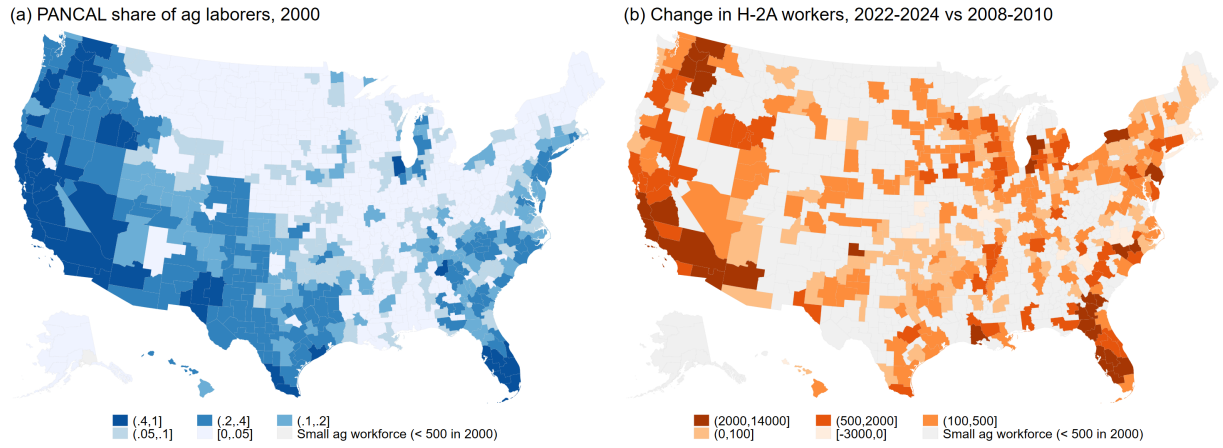


FIGURE 3. Baseline Exposure and the Growth of H-2A Employment, by Commuting Zone

Notes: Both panels show the 740 commuting zones in the estimation sample. Panel (a) shades zones by the PANCAL share of agricultural laborers in 2000, in round classes; the 10 percent boundary matches the high-exposure definition used elsewhere in the paper. Panel (b) shades zones by the change in certified H-2A workers, 2022–2024 average minus 2008–2010 average. Its class boundaries are zero, separating growth from no growth, and round values chosen so that the classes above zero hold 1.5, 11.8, 21.2, and 66.4 percent of the national increase, from lightest to darkest. Zones with fewer than 500 agricultural workers at baseline are left unshaded in both panels; the shaded zones carry 84.3 percent of year-2000 agricultural employment. *Sources:* American Community Survey via IPUMS; Census 2000; Department of Labor OFLC disclosure files; 1990 commuting zone boundaries.

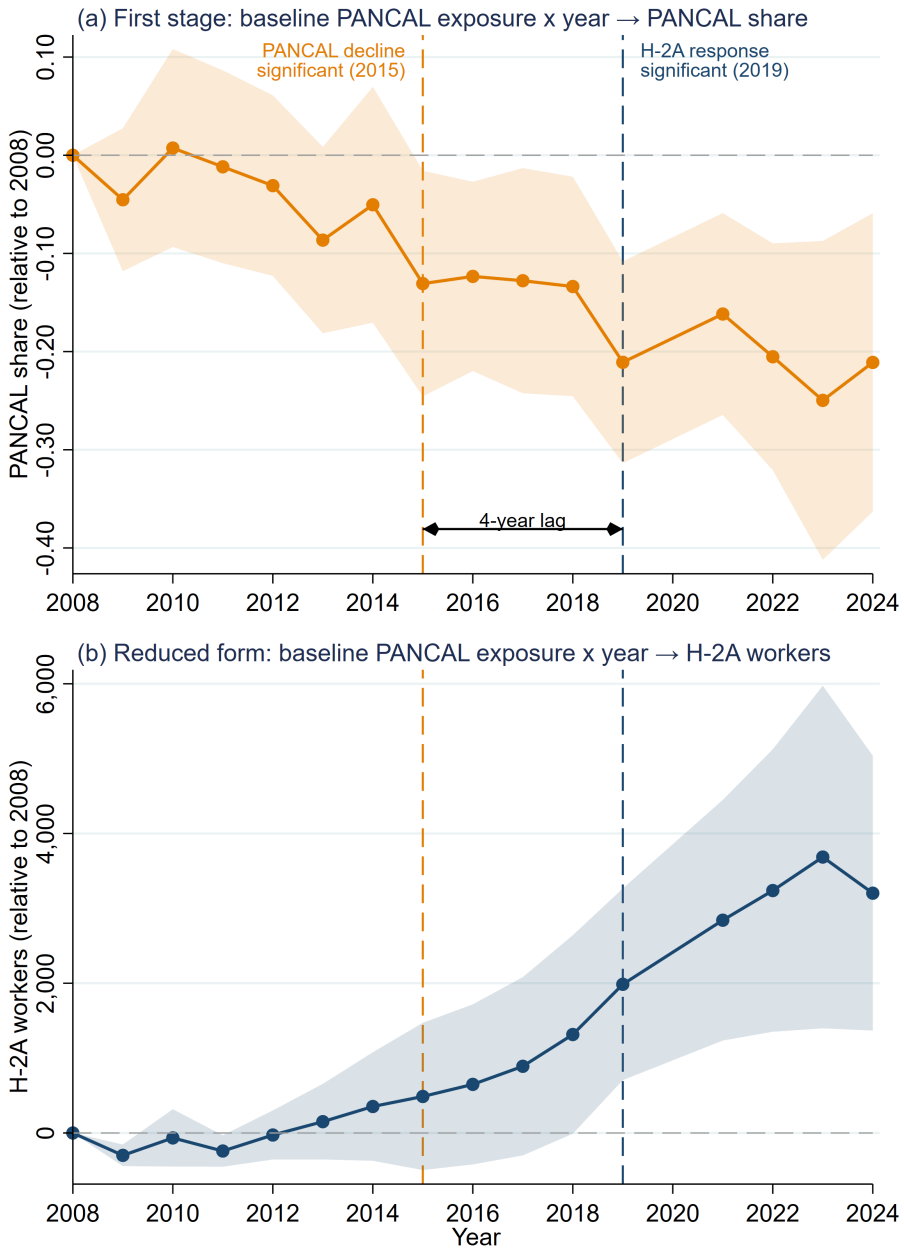


FIGURE 4. Event Studies of Baseline PANCAL Exposure

Notes: Coefficients from equation (5), with the PANCAL share as the outcome in panel (a) and H-2A workers in panel (b), estimated on the 740 commuting zones in the main sample with commuting zone and year fixed effects and vegetable share × year controls, weighted by year-2000 agricultural employment. There is no coefficient for 2020 because the ACS did not release standard one-year estimates for that year. Shaded bands are 95 percent confidence intervals clustered at the commuting zone level. Dashed vertical lines mark the first year in which the coefficient sequence is significant at the 5 percent level. *Sources:* American Community Survey via IPUMS; Department of Labor OFLC disclosure files.

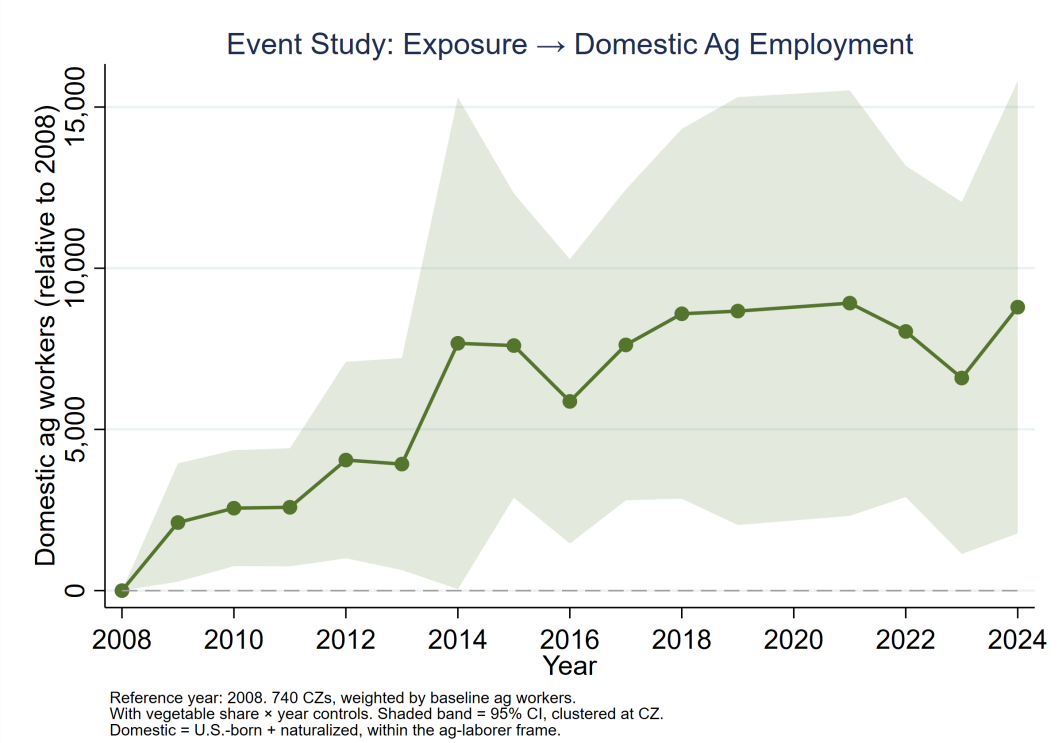


FIGURE 5. Event Study of Domestic Agricultural Employment on Baseline PANCAL Exposure

Notes: Each point is the coefficient on the year-2000 PANCAL share interacted with a year dummy, with domestic agricultural employment (U.S. citizens, born or naturalized) as the outcome and 2008 as the reference year. 740 commuting zones, weighted by year-2000 agricultural employment, with vegetable share × year controls. Shaded band is the 95 percent confidence interval clustered by commuting zone. The joint test that the 2009–2012 coefficients equal zero rejects at the 1 percent level.

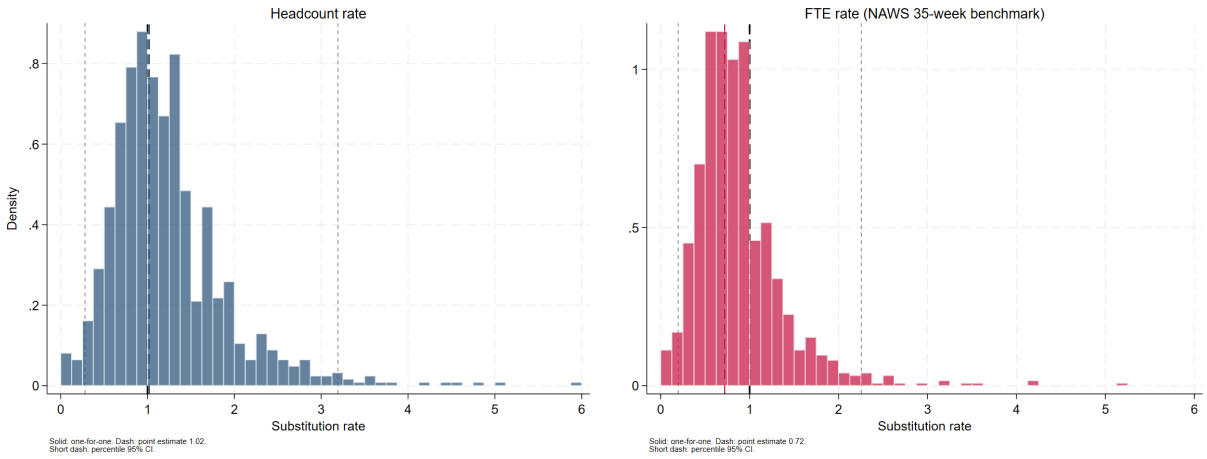


FIGURE 6. Bootstrap Distribution of the Substitution Rate

Notes: Histograms of the fixed-denominator substitution rate across 999 commuting-zone block bootstrap replications of Table 2, column (2), in headcount (left) and NAWS full-time-equivalent (right) units. The solid vertical line marks one-for-one replacement, the dashed line the full-sample point estimate, and the short-dashed lines the percentile 95 percent interval. Replications with a rate above 6 are not shown. See Table 5 and Appendix D.

TABLE 1. Summary Statistics

	(1) Full sample	(2) Low exposure (below median)	(3) High exposure (at or above median)
Panel A. Outcome and main regressor			
H-2A workers (count)	257.7	105.9	409.4
	(916.2)	(299.4)	(1,242.3)
PANCAL share (percent)	19.6	8.2	30.8
	(22.8)	(13.9)	(24.3)
Panel B. Agricultural employment and crop mix			
Total agricultural laborers	995.6	436.4	1,554.8
	(3,317.7)	(538.7)	(4,593.6)
Vegetable cropland share (percent)	1.0	0.3	1.8
	(3.4)	(0.9)	(4.6)
Panel C. Demographics and labor market			
Working-age population	280,794	128,718	432,870
	(779,586)	(281,307)	(1,044,134)
Hispanic share (percent)	11.4	4.5	18.3
	(14.7)	(3.7)	(18.0)
Employment rate (percent)	68.1	69.2	66.9
	(7.0)	(7.6)	(6.2)
Panel D. Geography			
Distance to U.S.-Mexico border (km)	1,503	1,726	1,281
	(763)	(716)	(744)
Panel E. Baseline (year-2000) shares, percent			
PANCAL share, 2000	12.0	1.4	22.7
	(14.7)	(1.8)	(14.2)
Non-agricultural settled non-citizen share, 2000	3.3	1.3	5.4
	(4.4)	(1.2)	(5.3)
Agricultural laborers, regression-weighted mean	9,062.5	—	—
Agricultural laborers, first-stage weighted mean ($\bar{N}^*$)	15,426.0	—	—
Observations (commuting zone-years)	11,840	5,920	5,920
Commuting zones	740	370	370

Notes: Panel means with standard deviations in parentheses; all shares and rates in percent. Column (1) is the full panel, 740 commuting zones over 2008 to 2024 excluding 2020; cells with zero agricultural employment have a missing PANCAL share and are dropped from estimation. Columns (2) and (3) split commuting zones at the median year-2000 PANCAL share of 6.0 percent. H-2A workers are certified positions, full and partial certifications. The full-sample mean of the 2000 PANCAL share, 12.0 percent, is below the pooled PANCAL mean in Panel A because the share rose between 2000 and the start of the panel: the unweighted mean is 21.2 percent in 2008–2010 and 17.0 percent in 2022–2024. Weighted by the 2000 agricultural workforce, the 2000 share is 30.1 percent and the panel share falls from 41.3 to 33.4 percent between the first and last three panel years; weighted means exceed unweighted means because farm labor is concentrated in high-share zones. The regression-weighted mean of agricultural laborers weights by the 2000 workforce; the first-stage weighted mean $\bar{N}^*$ is the denominator of the substitution rate (Section 4.2).

TABLE 2. The Effect of the Settled Non-Citizen Share on H-2A Employment

	(1) OLS	(2) Two-stage least squares	(3)	(4)
Excluded instruments	—	Budget	Encounters	Both
Panel A. Second stage. Dependent variable: H-2A workers				
PANCAL share (percent)	−2.11	−156.7	−226.5	−178.9
	(1.61)	(61.3)	(65.7)	(60.7)
Substitution rate (headcount)	—	1.02	1.64	1.20
Substitution rate (FTE, NAWS)	—	0.72	1.16	0.85
Panel B. First stage. Dependent variable: PANCAL share				
Border Patrol budget Bartik	—	−0.356	—	−0.304
		(0.093)		(0.092)
Agricultural-season encounter Bartik	—	—	−0.069	−0.047
			(0.012)	(0.010)
Kleibergen–Paap F-statistic	—	14.82	32.45	16.31
Panel C. Reduced form. Dependent variable: H-2A workers				
Border Patrol budget Bartik	—	55.8	—	41.9
		(16.1)		(13.1)
Agricultural-season encounter Bartik	—	—	15.5	12.6
			(4.0)	(3.3)
Anderson–Rubin p-value	—	< 0.001	< 0.001	< 0.001
Hansen J (p-value)	—	—	—	0.167
Observations	11,672	11,672	11,672	11,672

Notes: All regressions include commuting zone fixed effects, year fixed effects, and vegetable share × year interactions, and are weighted by year-2000 commuting zone agricultural employment. Standard errors, clustered at the commuting zone level (740 clusters), in parentheses. Panels B and C are estimated on the column (2)–(4) sample. In column (4), the reduced-form coefficients are from a single regression including both instruments; they are smaller than their single-instrument counterparts in columns (2) and (3) because the two Bartiks share the same exposure shares. The raw correlation between the two Bartiks is 0.997, but after fixed effects and controls the correlation of their residual variation is 0.28. The substitution rate is $|\hat{\delta}|/(0.01 \times \bar{N}^*)$, where $\bar{N}^*$ is the average zone agricultural workforce under the column's first-stage weights (15,426, 13,814, and 14,913 workers in columns (2) to (4)). Section 4.6 reports bootstrap intervals. *Sources:* Department of Labor OFLC disclosure files; American Community Survey via IPUMS; DHS Budget-in-Brief; CBP encounter records.

TABLE 3. Decomposing the OLS–IV Gap: Measurement Error versus Endogeneity

	(1) Coefficient	(2) Share of gap (percent)
Panel A. Estimates. Dependent variable: H-2A workers		
OLS	−2.11	—
	(1.61)	
Errors-in-variables-corrected OLS ($\hat{R} = 0.180$)	−11.7	—
	(9.4)	
Two-stage least squares (Border Patrol budget instrument)	−156.7	—
	(61.3)	
Panel B. Decomposition of the OLS–IV gap		
Total gap (IV minus OLS)	−154.6	100.0
Attributable to measurement error	−9.6	6.2
Attributable to endogeneity	−145.0	93.8
Panel C. Sensitivity of the corrected OLS to the reliability ratio		
Reliability $R = 0.10$	−21.1	12.3
Reliability $\hat{R} = 0.180$ (estimated)	−11.7	6.2
Reliability $R = 0.25$	−8.4	4.1
Reliability $R = 0.35$	−6.0	2.5

Notes: The reliability ratio R is the share of within-commuting-zone variance in the measured PANCAL share attributable to the true share rather than to sampling noise; across bootstrap replications $\hat{R}$ has mean 0.172 and standard deviation 0.019. The standard error of the corrected OLS estimate is from a block bootstrap with 999 replications resampling commuting zones, re-estimating both the coefficient and $\hat{R}$ in each. In Panel B the measurement error component is the corrected OLS estimate minus OLS; the endogeneity component is the IV estimate in Panel A minus the corrected OLS estimate. In Panel C, column (2) reports the share of the total gap that measurement error would explain at each value of R. OLS and IV specifications are as in Table 2.

TABLE 4. Share of H-2A Growth Explained by the PANCAL Decline

Instrument	Rate	Predicted ΔH-2A	Share explained (percent)
Border Patrol budget	1.02	85,243	28.5
Agricultural-season encounters	1.64	137,584	46.0
Both	1.20	100,683	33.7

Notes: The rate is $|\hat{\delta}|/(0.01 \times \bar{N}^*)$ from Table 2, columns (2)-(4). Predicted ΔH-2A is the rate times the decline in the settled non-citizen agricultural workforce, 83,912 workers, summed over the 735 commuting zones observed in both windows; the share divides by the observed H-2A growth of 298,923 over the same zones. Windows are the 2008–2010 and 2022–2024 averages.

TABLE 5. Bootstrap inference for the substitution rate

	Point	Bootstrap SE	Percentile 95 percent interval
Panel A. By unit			
Headcount	1.02	1.04	[0.28, 3.19]
FTE, NAWS 35-week	0.72	0.73	[0.20, 2.26]
FTE, 12-month	0.48	0.49	[0.13, 1.52]
Panel B. Headcount, by first-stage strength of the replication			
All replications (999)			[0.28, 3.19]
KP $F > 5$ (900)			[0.27, 2.30]
KP $F > 10$ (712)			[0.27, 1.95]

Notes: 999 commuting-zone block bootstrap replications of the benchmark specification. The rate divides $|\hat{\delta}|$ by $0.01 \times \bar{N}^*$, with $\bar{N}^*$ held at its full-sample value of 15,426, the average zone workforce under the first-stage weights. Appendix D reports the version that recomputes $\bar{N}^*$ in each replication. FTE conversions use the NAWS 35-week work year (factor 0.707) and a 12-month year (factor 0.476).

TABLE 6. Alternative Exposure Measures

Panel A. Horse races, reduced form. Dependent variable: H-2A workers

	Non-agricultural share	Border distance
Agricultural Bartik (B^{ag})	74.2	74.1
	(26.3)	(23.3)
Alternative Bartik	−82.3	−8.3
	(84.1)	(5.5)

Panel B. Two-stage least squares. Dependent variable: H-2A workers

	Non-agricultural share as control	Border distance as instrument
PANCAL share (percent)	−187.0	−28.0
	(81.6)	(59.8)
Kleibergen-Paap F	12.65	4.67

Notes: The agricultural Bartik is the main instrument, the year-2000 PANCAL share interacted with the Border Patrol budget. The alternative Bartik in Panel A, column (1), interacts the year-2000 settled non-citizen share of low-education non-agricultural employment (high school or less, aged 16 to 64) with the same shifter; the two baseline shares have a correlation of 0.70 across commuting zones. In column (2) it interacts negative log distance to the Mexican border with the same shifter. Each column of Panel A enters the agricultural Bartik and the alternative Bartik in one reduced form, equation (6). Panel B, column (1) adds the contemporaneous non-agricultural settled share as a control to Table 2, column (2), which gives −156.7 (61.3) without it; column (2) instruments the PANCAL share with the distance Bartik alone. Specification, weights, and clustering as in Table 2. Standard errors clustered by commuting zone in parentheses.

TABLE 7. The Livestock Placebo

Panel A. First stage. Dependent variable: livestock PANCAL share (percent)		
	Border Patrol budget	Encounters
Livestock Bartik (B^{lvstk})	−0.57	−0.15
	(0.20)	(0.04)
Panel B. Horse race, equation (7). Dependent variable: H-2A workers		
Crop Bartik (B^{crop})	92.2	
	(29.4)	
Livestock Bartik (B^{lvstk})	−60.6	
	(34.2)	

Notes: Livestock is IPUMS 1990 industry code 11 (livestock production, including dairy, cattle, hogs, and poultry), with the same occupation and settled non-citizen filters as PANCAL. The crop and livestock Bartiks interact the year-2000 settled non-citizen shares of crop and livestock employment with the enforcement shifter; Panel B uses the Border Patrol budget shifter. Livestock cells in the ACS are thin, so the first stage in Panel A is read as a sign and significance check rather than a precise magnitude. Specification, weights, and clustering as in Table 2. Standard errors clustered by commuting zone in parentheses.

TABLE 8. Domestic Agricultural Employment and the Substitution Estimate

	Border Patrol budget	Encounters	Both
Panel A. Domestic employment as the outcome			
PANCAL share (percent)	−381.1	−180.7	−317.3
	(141.5)	(86.2)	(114.6)
Kleibergen-Paap F	14.82	32.45	16.31
Hansen J (p-value)	—	—	0.105
U.S.-born employment as the outcome	—	—	−243.6
			(91.0)
Pre-period joint test, event study (p-value)	0.003 (U.S.-born only: 0.004)		
Panel B. H-2A workers as the outcome, with pre-period domestic growth × year controls			
PANCAL share (percent)	−181.0	−189.1	−184.8
	(97.4)	(72.5)	(81.9)
Kleibergen-Paap F	9.50	27.53	14.04
Hansen J (p-value)	—	—	0.89
Same specification, no control	−156.7	−226.5	−178.9

Notes: Domestic employment is agricultural laborers who are U.S. citizens, born or naturalized. Panel A replaces H-2A workers with domestic employment as the outcome in the specification of Table 2; the U.S.-born row restricts to workers born in the United States. The pre-period test is the joint test that the 2009-2012 coefficients in the domestic event study (Figure 5) equal zero. Panel B adds to the main specification each commuting zone's 2008-2012 domestic employment growth interacted with year dummies. Specification, weights, and clustering as in Table 2.

TABLE 9. Alternative Instruments

	$\hat{\delta}$	Kleibergen-Paap F	Hansen J (p-value)
Panel A. Smooth national shifters in place of the budget			
Consumer price index	−182.8 (63.7)	19.00	—
Federal health spending	−175.4 (59.0)	22.35	—
Linear trend	−168.7 (57.5)	22.36	—
Agricultural-season encounters (non-smooth)	−226.5 (65.7)	32.45	—
Panel B. Supply-shift instruments (baseline share × national PANCAL *decline)*			
Leave-one-out national share, alone	−222.4 (83.9)	16.10	—
National share, alone	−179.9 (60.5)	24.43	—
Leave-one-out share + budget	−177.2 (67.3)	8.40	0.08
Leave-one-out share + encounters	−223.9 (75.5)	17.18	0.92
All three instruments	−180.5 (61.6)	11.93	0.21

Notes: Each row is the specification of Table 2 with the instrument set named. In Panel A the year-2000 PANCAL share is interacted with the log of the named national series in place of the Border Patrol budget. In Panel B the supply-shift instrument interacts the year-2000 PANCAL share with the log of the national PANCAL share, computed excluding the commuting zone's own workers (leave-one-out) or including them. Standard errors clustered by commuting zone in parentheses.

TABLE 10. Rotemberg Weights: Where the Identification Comes From

	Budget	Encounters	Both
Weight in the overidentified estimate	0.68	0.32	—
Weights by period (sum of year weights)			
2008-2017	−23.9	−12.0	−20.1
2018-2024	24.9	13.0	21.1
Three largest year weights	2023, 24, 19	2023, 24, 22	2023, 24, 19
Weights by commuting zone			
California share of identifying weight	0.49	0.37	0.45
Zones with negative weight (of 740)	217	267	227

Notes: Rotemberg decomposition following Goldsmith-Pinkham et al. (2020). The instrument weights are the shares of the overidentified estimate attributable to each just-identified estimate; the weighted just-identified estimates reproduce −156.7, −226.5, and −178.9 exactly. Year weights sum to one within each column; a negative sum over 2008-2017 reflects a national shifter that had not yet moved. The California share is the sum of weights on California commuting zones.

TABLE 11. Controls Progression

	Border Patrol budget		Both instruments	
Controls interacted with year	$\hat{\delta}$	KP F	$\hat{\delta}$	KP F
None	−414.8 (178.5)	11.64	−467.6 (178.7)	17.53
Labor-intensive crop share	−50.7 (96.1)	7.24	−60.3 (107.7)	4.53
Vegetable, orchard, and tobacco shares	−118.1 (84.1)	6.85	−136.7 (88.8)	4.53
Vegetable share (main specification)	−156.7 (61.3)	14.82	−178.9 (60.7)	16.31

Notes: Each row is the specification of Table 2 with the control set named in the first column, all fixed at their 2002 Census of Agriculture values and interacted with year dummies. The labor-intensive crop share is the combined acreage of vegetables, orchards, and berries as a share of cropland; the other rows use the single-crop shares of cropland. Hansen J p-values for the both-instruments column, top to bottom: 0.16, 0.50, 0.16, 0.17. Standard errors clustered by commuting zone in parentheses.

TABLE 12. Specification and Sample Robustness

	$\hat{\delta}$	KP F	J (p)	Rate
Panel A. Levels, both instruments				
Main specification	−178.9 (60.7)	16.31	0.17	1.20
California and weighting				
Excluding California	−346.2 (162.8)	5.55	0.57	—
Excluding three largest California zones	−182.7 (103.5)	8.94	0.93	—
Unweighted	−65.9 (21.7)	28.99	0.52	1.94
Dropping top 1% of H-2A outcomes	−168.2 (94.7)	4.12	0.23	—
Sample and timing				
Crop agriculture only	−211.6 (78.3)	13.06	0.39	2.01
Recent-arrival share control (migration)	−172.9 (56.5)	16.58	0.08	1.06
Recent-arrival share control (years in U.S.)	−181.3 (62.4)	15.34	0.30	1.29
Instruments lagged one year	−179.9 (60.7)	12.02	—	1.41
PANCAL share lagged one year	−193.1 (65.0)	15.35	—	1.29
Wider definitions of the main regressor				
Mexican-born share	−214.3 (77.1)	9.47	0.04	—
Foreign-born share	−216.2 (75.3)	8.55	0.04	—
Panel halves				
2008-2017 only	−69.1 (59.2)	2.43	0.73	—
2018-2024 only	−325.4 (204.3)	1.90	—	—
Panel B. Proportional outcomes, both instruments				
asinh(H-2A workers)	−0.089 (0.038)	16.31	0.33	0.59
ln(H-2A workers), zeros dropped	−0.124 (0.037)	14.79	0.18	0.90

Notes: Each row is the overidentified specification of Table 2, column (4), with the change named. Standard errors clustered by commuting zone in parentheses. Rates are $|\hat{\delta}|/(0.01 \times \bar{N}^*)$ with $\bar{N}^*$ the average zone employment under the first-stage weights of the row's own specification; rates are not reported for rows with a Kleibergen-Paap F below 10, where the first-stage weights that define $\bar{N}^*$ are unstable (Appendix D). In Panel B the implied rate evaluates the proportional effect at the weighted sample-mean H-2A level (987 in the full sample; 1,080 among positive cells) and divides by the specification's $\bar{N}^*$; it is an accounting conversion, not a second estimate. The log specification drops 2,531 of 11,840 commuting-zone-years with zero H-2A workers, 480 of them in commuting zones that never use the program.

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A Data Construction

A.1 H-2A certified positions

The outcome is the number of H-2A positions certified in each commuting zone and year, built from the Department of Labor's Office of Foreign Labor Certification disclosure files for fiscal years 2008 to 2024. Each record is an employer application with the number of workers requested and certified, the case status, the worksite location, and the requested employment dates. Applications are deduplicated by case number, so multi-worksite filings contribute their certified positions once. Worksites are assigned to counties using the county field where the files record one (fiscal 2017 onward), ZIP codes for 2015 and 2016, and city names with manual assignment for 2008 to 2014; counties are then mapped to 1990 commuting zones with the Dorn crosswalk. Geocoding coverage is 97 to 99.6 percent of positions in the county-field years and 93 to 97.5 percent in the city-name years, so a larger share of early-year positions is unassigned to a commuting zone.

An application can be certified in full or in part. A partial certification approves fewer workers than the employer requested, and on those records the certified worker count is the reduced approved number. The outcome includes both, consistent with OFLC's published totals of certified positions. The constructed series is within 3 percent of the published figure in every such year except fiscal 2008, where it exceeds the published figure by 6 percent, or about 5,100 positions. That excess is close to the number of positions on partial certifications in the fiscal 2008 file, which suggests that the published fiscal 2008 aggregate excludes partial determinations, unlike the published totals for later years. OFLC's published figures include Puerto Rico; the commuting-zone panel excludes it.

A.2 The settled non-citizen share

The main regressor is the settled non-citizen share of agricultural laborers, PANCAL, built from the American Community Survey one-year files via IPUMS for 2008 to 2024, excluding 2020, when the ACS did not release standard one-year estimates. Agricultural laborers are employed persons aged 16 to 64 in crop production, livestock production, or agricultural services (1990 industry codes 10, 11, and 20) in farm-worker, nursery-worker, or supervisor occupations (1990 occupation codes 479, 484, and 488). A laborer is settled non-citizen if not a U.S. citizen and not living abroad one year before the survey. The share is the number of settled non-citizen laborers over all agricultural laborers, computed with person weights and expressed in percent. Recently arrived non-citizens, those who lived abroad one year earlier, are excluded from the numerator by construction; Section 5.5 and Appendix F show why

that exclusion matters.

The exposure share $s_{i,2000}$ applies the same industry, occupation, and citizenship filters to the 2000 decennial census 5 percent sample. The one-year migration filter cannot be applied there, because the 2000 census records residence five years earlier rather than one year earlier; I use years in the United States greater than one in its place, which the census does record.

A.3 National enforcement series

The budget shifter is the real annual Border Patrol budget in 2024 dollars, assembled from Department of Homeland Security Budget-in-Brief documents, Congressional Research Service report RL32562, and appropriations records, and deflated by the CPI-U. The encounter shifter is the number of U.S.-Mexico border encounters recorded by Customs and Border Protection during March through September, the months of peak agricultural hiring. Both enter the instruments in logs. The comparison series used in Section 5.4 are the CPI-U and real federal health spending, from the Bureau of Labor Statistics and the Bureau of Economic Analysis.

A.4 Crop controls

Crop composition is from the 2002 Census of Agriculture, accessed through USDA Quick Stats, at the county level: total cropland and acreage in vegetables, orchards, berries, and tobacco. Counties are aggregated to commuting zones with the same crosswalk and each category is expressed as a share of cropland. The main specification uses the vegetable share interacted with year dummies; Section 5.5 reports the alternatives.

TABLE A1. Variable Definitions and Sources

Variable	Years	Definition	Source
Outcome			
H-2A workers	2008-2024	Certified H-2A positions per commuting zone-year, full and partial certifications	DOL OFLC disclosure files
Endogenous variable			
PANCAL share	2008-2024	Non-citizen agricultural laborers who did not live abroad in the prior year, as a share of all agricultural laborers (percent)	ACS one-year via IPUMS
Instrument: exposure share			
PANCAL share, 2000	2000	Same construction, with years in the U.S. greater than one replacing the migration filter	2000 census 5 percent sample via IPUMS
Instrument: national shifters			
Border Patrol budget	2008-2024	Log real annual budget, 2024 dollars	DHS Budget-in-Brief; CRS RL32562; CPI-U
Ag-season encounters	2008-2024	Log March-September U.S.-Mexico border encounters	CBP encounter records
Controls			
Crop shares	2002	Vegetable, orchard, berry, and tobacco acreage as shares of cropland	USDA Census of Agriculture
Weights			
Agricultural employment, 2000	2000	Agricultural laborers per commuting zone	2000 census via IPUMS

B H-2A Contract Duration and the Labor-Time Conversion

The full-time-equivalent substitution rates in Section 4.6 convert headcounts to labor time using the average length of an H-2A contract. Figure B1 plots the worker-weighted mean contract duration by year from the certification data, where duration is the requested employment end date minus the start date on each application. Contracts ran between 6.4 and 6.8 months through 2011 and have been close to 5.5 months since 2013, the period in which the settled workforce declines and H-2A employment grows; the worker-weighted mean over 2008 to 2024 is 5.69 months. The NAWS benchmark for settled workers is 35 weeks, or 8.08 months, of farm work per year, the average over the six survey cycles spanning 2007–2020. The two conversion factors are $5.69/8.08 = 0.705$ (NAWS benchmark) and $5.69/12 = 0.474$ (twelve-month benchmark).

B.1 National enforcement and comparison series

The instruments interact baseline exposure with a national series. Figure B2 plots the series used in the paper and in the robustness checks of Section 5.4, each standardized to mean zero and standard deviation one. The Border Patrol budget, the consumer price index, and federal health spending trend together over the period, which is why a smooth shifter interacted with baseline exposure identifies much the same variation regardless of which series is used. The agricultural-season encounter series follows a different path.

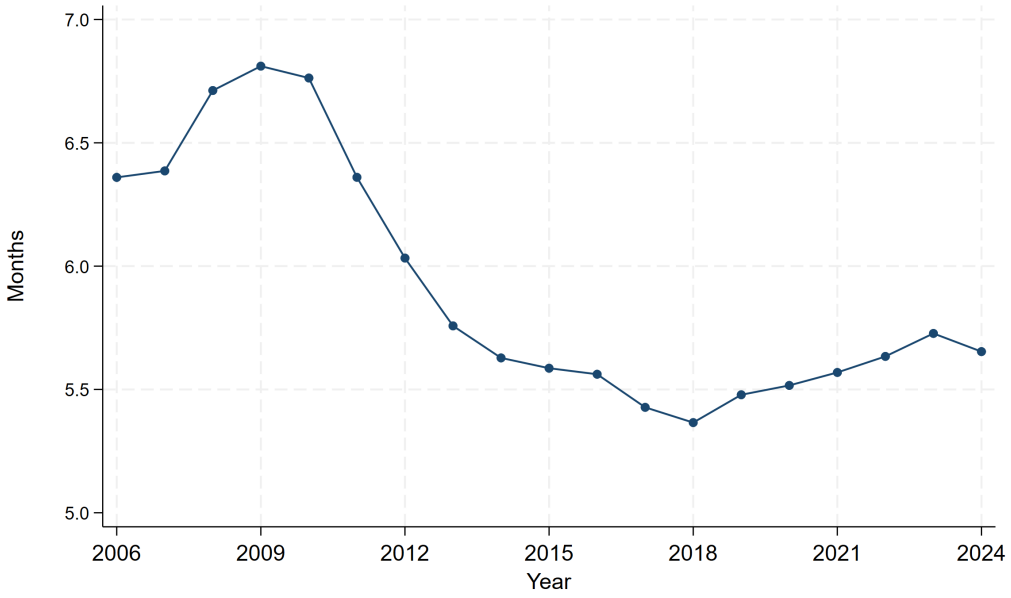


FIGURE B1. Average H-2A Contract Duration by Year

Notes: Worker-weighted mean length of certified H-2A contracts, in months, by fiscal year of certification. Contract length is the requested end date minus the requested start date on each employer application, weighted by the number of workers certified. Applications with missing, non-positive, or implausibly long durations (over 37 months) are dropped. *Sources:* Department of Labor OFLC H-2A disclosure files, 2006–2024.

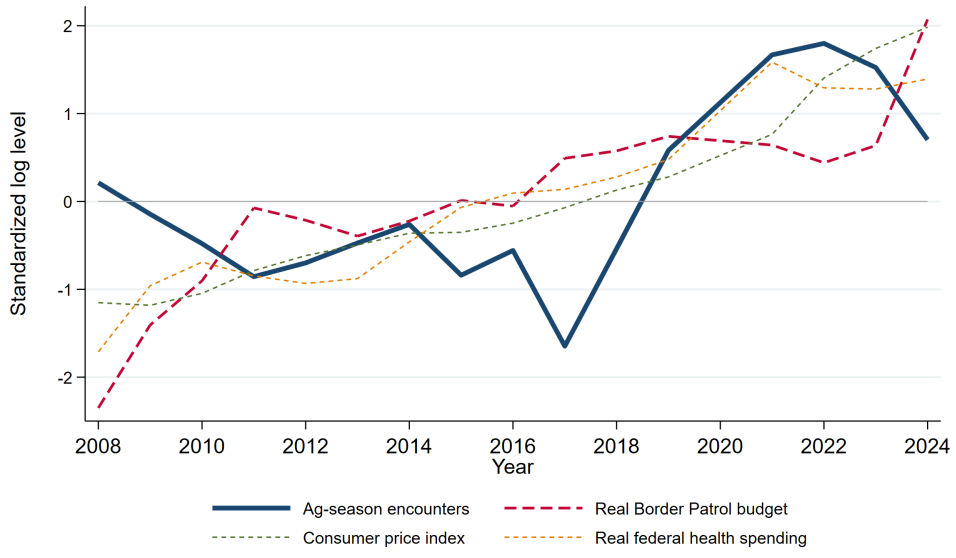


FIGURE B2. National Shifter Series Used to Construct the Instruments

Notes: Each series is the log of the national series, standardized to mean zero and standard deviation one over 2008 to 2024. A linear trend, also used in Section 5.4, is not plotted. *Sources:* DHS Budget-in-Brief; CBP encounter records; Bureau of Labor Statistics; Bureau of Economic Analysis.

C Lagged Instruments

The event studies date the decline of the settled share from 2015 and the rise of H-2A hiring from 2019. The four-year gap is the distance between two significance dates, not an estimated lag. This appendix asks whether the design can estimate the lag directly, by regressing H-2A workers in year t on the PANCAL share in year $t-k$, instrumented with the budget Bartik in year $t-k$, for $k = 1$ to 4. Lagging shortens the sample, so each lagged estimate is paired with the contemporaneous estimate on the same observations. Specification, weights, and clustering are as in the main text. Lags that fall on 2020 are missing because the ACS released no standard 2020 estimates.

Table C1 reports the results. The lagged and contemporaneous estimates are statistically indistinguishable at every lag. Both grow with k because the sample shifts toward later years, where the effect is larger, not because the lag matters. A distributed-lag version that enters the share at t and at $t-4$ together cannot separate them: the joint first-stage F is 0.3, neither coefficient is individually significant, and only their sum is identified (-191.1, $p = 0.004$). The reason is the shifter. The budget grows smoothly in logs, so the lagged instrument equals the current instrument minus a zone constant, which the fixed effects absorb; after fixed effects, the lagged and current instruments carry nearly the same variation. The design therefore does not identify the lag length, which is why the paper reports the event-study dates as description and makes no claim about adjustment speed.

TABLE C1. Lagged Instruments, Budget Bartik

Lag k	$\hat{\delta}$ at $t-k$	KP F	$\hat{\delta}$ at t, same sample	KP F	Observations
1	−174.1 (70.1)	13.7	−172.3 (67.5)	15.0	10,108
2	−180.2 (69.6)	16.5	−169.4 (66.2)	15.8	9,406
3	−200.3 (76.4)	13.4	−199.7 (85.9)	11.7	8,706
4	−215.9 (87.7)	10.9	−191.1 (69.7)	16.0	7,962

Notes: Each row instruments the PANCAL share at $t-k$ with the budget Bartik at $t-k$ (columns 2-3) and, on the same observations, the share at t with the Bartik at t (columns 4-5). The full-sample contemporaneous estimate is −156.7 (61.3). Standard errors clustered by commuting zone in parentheses. Specification and weights as in Table 2.

D The Substitution Rate: Tail Diagnostics

Section 4.6 reports that the bootstrap distribution of the substitution rate is right skewed. This appendix documents where the tail comes from. All results are for the benchmark specification, Table 2, column (2), and the commuting-zone block bootstrap with 999 replications described

there.

D.1 Weak draws

Each replication's first-stage strength was recorded. Across the 999 draws the median Kleibergen-Paap F is 15.3, close to the full-sample 14.8, but 287 draws fall below 10: resampling produces weak-instrument draws, and weak instruments inflate the coefficient. Conditioning on first-stage strength trims the tail without moving the low side: with the denominator held fixed, the 95 percent interval is $[0.28, 3.19]$ over all draws, $[0.27, 2.30]$ for draws with F above 5, and $[0.27, 1.95]$ for draws with F above 10, while the probability of a rate below one moves from 42 to 53 percent.

D.2 Recomputing the denominator in each replication

The featured interval holds $\bar{N}^*$ at its full-sample value. $\bar{N}^*$ is itself estimated, so this subsection reports a bootstrap in which each replication recomputes its own first-stage weights and its own $\bar{N}^*$. The center matches the featured interval: the median headcount rate is 1.32 and the median labor-time rate is 0.93. The tails do not. The 95 percent headcount interval is $[0.37, 23.39]$, and in 110 of 999 replications the signed weight sum is degenerate and $\bar{N}^*$ is undefined. The cause is the same geographic concentration documented above. Replications that underweight the California core weaken the first stage, inflate $|\hat{\delta}|$, and destabilize the weights, so the numerator rises while the denominator collapses; across replications the correlation between $\hat{\delta}$ and $\bar{N}^*$ is 0.50, and the smallest computable $\bar{N}^*$ is 22 workers. Conditioning on replications with a first-stage F above 10 brings the interval to $[0.32, 6.90]$. The instability therefore reflects the resampling, not the full-sample estimate, whose $\bar{N}^*$ is computed from a strong first stage ($F = 14.8$).

D.3 No single zone is influential

A leave-one-zone-out jackknife re-estimates the benchmark and its $\bar{N}^*$ 740 times, dropping one zone each time. The median rate is 1.02, and 739 of the 740 replications lie within $[0.85, 1.86]$. The exception is dropping the Salinas commuting zone, which cuts $\bar{N}^*$ to 2,182 workers, lowers the first-stage F to 7.7, and pushes the rate to 9.6, the same weak-first-stage mechanism as in the bootstrap. Holding $\bar{N}^*$ at its full-sample value, all 740 replications lie within $[0.84, 1.36]$.

D.4 Which places

Correlating each zone's draw count across replications with the replication's rate identifies the geography. The zones whose presence holds the rate down, and whose joint omission produces the tail, are the California core: the Fresno-Tulare San Joaquin Valley zone, which alone carries 5.7 percent of national baseline agricultural employment, greater Los Angeles, the Salinas Valley, Kern County, and Merced-Stanislaus. Their profile is high exposure, with baseline PANCAL shares of 60 to 72 percent, very large weight, and low H-2A intensity, 0.04 to 0.14 H-2A workers per agricultural worker, with the Salinas Valley higher at 0.50. The zones whose presence pushes the rate up are the H-2A-intensive regions: the Yakima Valley, the Columbia Basin, and Adams-Grant in Washington, and Palm Beach and Polk-Highlands in Florida, with 0.8 to 2.2 H-2A workers per agricultural worker, an order of magnitude above the California core's.

The mechanism, in one sentence: the national rate is an average over places with very different H-2A intensity. California, where the settled workforce is concentrated and H-2A use is comparatively light, anchors it near one; draws that underweight California thin the identifying variation and weaken the first stage, and let the Washington and Florida H-2A belts dominate. Two deterministic checks on the real data confirm this. Dropping the three California anchor zones outright leaves the coefficient essentially unchanged, −182.7 (103.5) against −178.9 (60.7) in the overidentified specification, while the identifying workforce shrinks from 14,913 to 3,953 workers and the first stage weakens to an F of 8.9, so no rate is reported; the tail is denominator geography, not coefficient instability. Dropping the top 1 percent of commuting-zone-year H-2A outcomes gives −168.2 (94.7), so the largest program cells are not driving the estimate either, though that check is weakly identified (first-stage F of 4.1) and is read as a point-estimate comparison only.

E ACS Coverage of the Settled Workforce

The ACS counts settled non-citizen workers where they live; the Census of Agriculture counts hired workers where they work, and reaches populations that household surveys miss. If the ACS undercounts the settled non-citizen workforce, then a given ACS decline understates the number of workers actually lost, and the headcount substitution rate overstates replacement. This appendix bounds the adjustment.

The bound applies the NAWS settled non-citizen share of hired crop workers to the Census of Agriculture count of workers employed 150 days or more, to obtain an implied settled non-citizen workforce; the ratio of the ACS PANCAL count to it is a coverage rate κ. In the fiscal 2021-2022 NAWS microdata, settled non-citizens are 57.7 percent of hired crop workers (two thirds of them unauthorized, which is the population subject to the enforcement instruments). The Census counts of workers employed 150 days or more are 980,114, 957,437, and 1,029,715 in 2012, 2017, and 2022, so the implied settled non-citizen workforce is about 565,000, 552,000, and 594,000, against ACS PANCAL counts of 355,000, 316,000, and 250,000. National coverage is therefore $\kappa = 0.63$ in 2012, 0.57 in 2017, and 0.42 in 2022. Table E1 reports the substitution rate under each value, assuming undercoverage is proportional across commuting zones. Under these values the headcount rate lies between 0.43 and 0.64. A proportional undercount scales the rate down, so it cannot push the estimate above the benchmark; the benchmark is the upper bound.

Three caveats make this an accounting bound rather than a measurement. The NAWS share is a national crop-worker average applied to an all-agriculture total. The 150-day count includes H-2A workers whose contracts exceed 150 days, which inflates the denominator after 2017 and makes κ understate coverage in the surge years. And 910 county-years are suppressed in the Census data and 13 counties do not match a commuting zone. At the commuting-zone level, coverage is far from uniform: the workforce-weighted κ matches the national figures, but the median zone has κ of 0.10, 0.08, and 0.04 in the three years, 22 to 35 percent of zones record no PANCAL workers at all, and κ rises steeply with baseline exposure, from below 0.1 in the least exposed tercile to 0.96, 0.89, and 0.65 in the most exposed. The cells that carry the identification are the well-measured ones, so the proportional adjustment overstates the correction that applies to them.

TABLE E1. Substitution Rate Under Alternative Units and Coverage Assumptions

	Headcount	FTE (35-week season)	FTE (12-month)
Benchmark (ACS as measured)	1.02	0.72	0.48
ACS coverage $\kappa = 0.63$ (2012)	0.64	0.45	0.30
ACS coverage $\kappa = 0.57$ (2017)	0.58	0.41	0.28
ACS coverage $\kappa = 0.42$ (2022)	0.43	0.30	0.20

Notes: The benchmark is Table 2, column (2), scaled by the first-stage weighted workforce $\bar{N}^*$ (Section 4.2). FTE factors: 35-week season, $5.71/8.08 = 0.707$; 12-month, $5.71/12 = 0.476$. κ is the ratio of ACS PANCAL workers to the implied settled non-citizen workforce, computed as Census of Agriculture hired workers employed 150 days or more times the NAWS settled non-citizen share of hired crop workers (57.7 percent, FY2021-2022 microdata). Adjusted rates equal the benchmark rate times κ and assume undercoverage is proportional across commuting zones. The 150-day count includes H-2A workers whose contracts exceed 150 days, which inflates the denominator after 2017 and understates coverage in those years.

F Measurement Diagnostics

The PANCAL measure excludes non-citizens who lived abroad in the prior year. Three diagnostics show that the exclusion insulates it from the program it is used to explain.

F.1 Group quarters

The ACS added group quarters, including labor camps, worker dormitories, and employer housing, in 2006, and this is where guestworkers would appear if the survey caught them. Figure F1 plots the share of agricultural laborers living in group quarters by group. Settled non-citizen laborers live in group quarters at a rate of 4.8 percent over 2006-2017 and 5.3 percent over 2018-2024, essentially flat. Recently arrived non-citizens do so at 28.4 percent before 2018 and 46.0 percent after, and in every year from 2021 through 2024 the share is between 48 and 63 percent. Domestic workers are at 1.9 percent throughout. The arrival margin absorbs the H-2A surge; the settled margin does not. The recent-arrival cells are thin, between 187 and 601 unweighted respondents per year, so the year-to-year series is noisy, but the era means are not.

F.2 Seasonality

Because the ACS samples throughout the year, a difference in seasonal work patterns between settled non-citizen and domestic laborers could enter the share measure. Monthly Current Population Survey data show that it does not. Figure F2 plots the seasonal index of agricultural laborers by group: non-citizen and domestic employment share the same profile, a July peak and March trough, with an amplitude of 29.8 index points for non-citizens and 32.8 for domestic workers over 2008-2024, and for non-citizens it is 28.4 in 2008-2017 against 33.9 in 2018-2024. The ACS's rolling sample does not interact differentially with the settled non-citizen measure.

F.3 Composition of the settled workforce

The National Agricultural Workers Survey is the only survey that records legal status. In the fiscal 2021-2022 microdata, settled non-citizens are 57.7 percent of hired crop workers, 66.8 percent of them unauthorized; unauthorized workers are 68.2 percent of all non-citizens, and 5.9 percent of foreign-born workers are in their first year in the United States. This is the population the PANCAL measure captures and the one subject to the enforcement instruments. NAWS weights are relative rather than population weights, so the survey yields shares, not counts.

F.4 The crop controls

The vegetable share of cropland is the crop control in the main specification. Two figures document why it is the composition that matters and what it removes.

Figure F3 plots the crop composition of certified H-2A workers over 2011 to 2019, the years for which the disclosure files record a crop. The vegetable share rises from 16 percent in 2011 to 32 percent in 2016 and remains above a quarter through 2019, while the general or unspecified category falls from 20 percent to under 10 percent before recovering. Berries rise from 3 to 9 percent; tobacco, apples, and nursery stock are flat. The program's growth therefore concentrated in vegetables, the crop group in which settled non-citizen workers were also most common. A commuting zone with vegetable acreage at baseline would have expanded H-2A hiring over this period regardless of what happened to its settled workforce, and the vegetable share interacted with year dummies absorbs that path.

Figure F4 shows the reduced-form event study with and without the vegetable controls. Without them, the coefficients after 2017 are roughly twice as large, peaking near 7,800 in 2022 against 4,000 with the controls; with them, they are the coefficients reported in Figure 4. The pre-period coefficients are close to zero in both versions, and the rise begins in the same years. The crop-demand confound therefore inflates the magnitude of the post-period response but does not create its timing or its pre-period pattern.

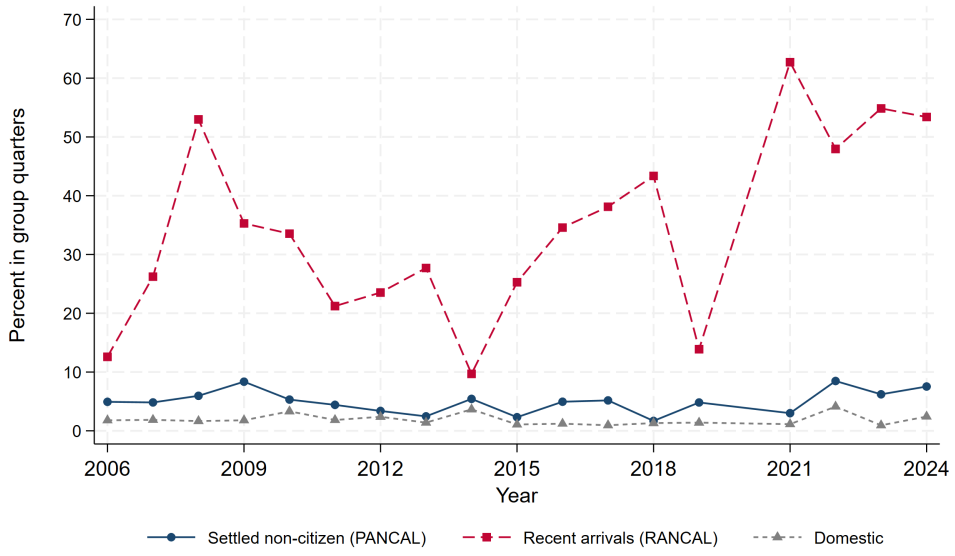


FIGURE F1. Share of Agricultural Laborers Living in Group Quarters, by Group

Notes: Person-weighted share of agricultural laborers in group quarters (IPUMS gq codes 3 and 4), by year and group. PANCAL is settled non-citizens; recent arrivals are non-citizens who lived abroad one year earlier; domestic is citizens. *Source:* ACS one-year files via IPUMS, 2006-2024.

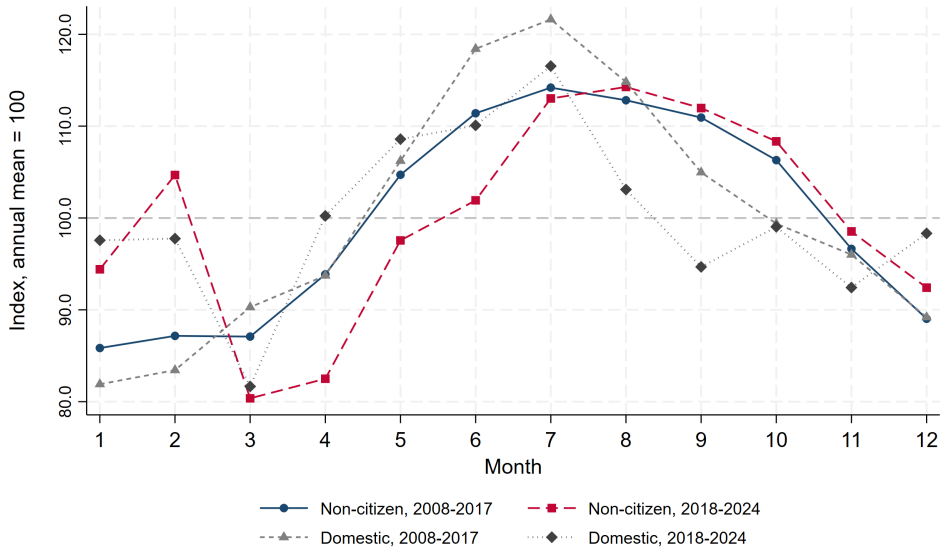


FIGURE F2. Seasonal Profile of Agricultural Laborers, CPS Monthly

Notes: Monthly employment of agricultural laborers indexed to the annual mean of 100, by citizenship and era. Same industry and occupation definitions as the ACS frame. Roughly 250 to 350 respondents per month. *Source:* CPS Basic Monthly files via IPUMS, 2008-2024.

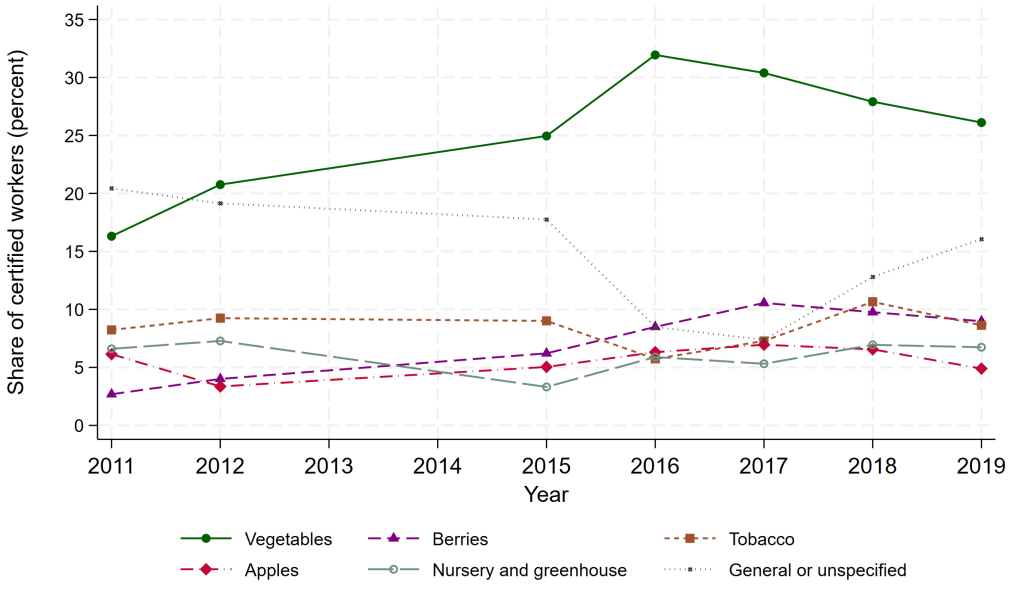


FIGURE F3. Crop Composition of Certified H-2A Workers

Notes: Worker-weighted shares of certified H-2A positions by crop, from the OFLC disclosure files. Crop information is unavailable for 2013 and 2014 and is not reported after 2019. Vegetables include melons, following the USDA-NASS classification. Only the six largest crop groups are shown; the general or unspecified category covers applications without a crop designation.

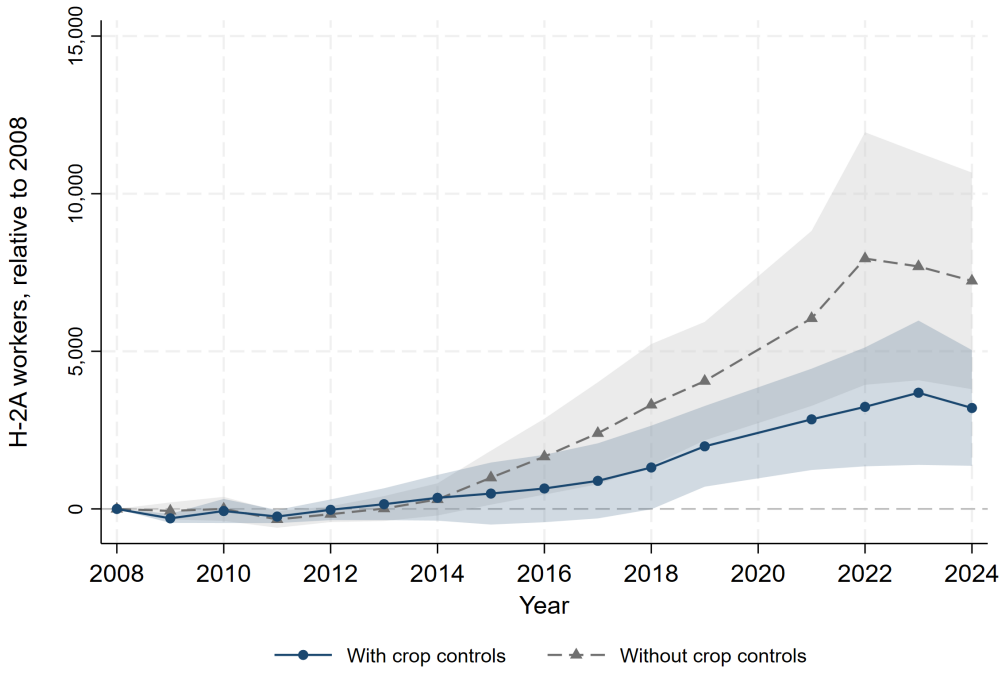


FIGURE F4. Reduced-Form Event Study With and Without Crop Controls

Notes: Coefficients on the year-2000 PANCAL share interacted with year dummies, with H-2A workers as the outcome and 2008 as the reference year, estimated with and without the vegetable share × year controls of the main specification. 740 commuting zones, weighted by year-2000 agricultural employment. Shaded bands are 95 percent confidence intervals clustered at the commuting zone level.